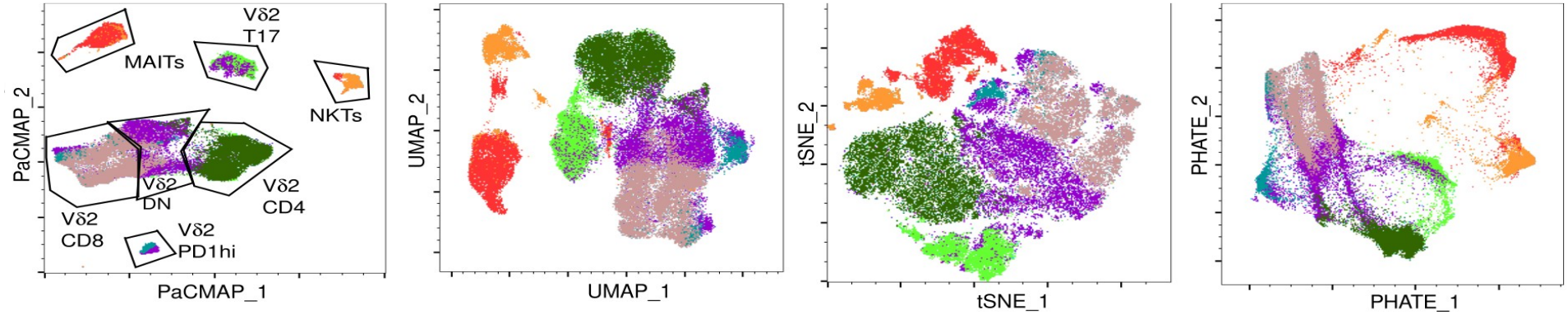
UNIVERSITY of MARYLAND
MARLENE AND STEWART GREENEBAUM
COMPREHENSIVE CANCER CENTER

ISAC RECOGNIZED
Shared Resource Laboratory



Towards ‘No Cell Left Behind’: open-source tool sets to leverage Spectral Flow Cytometry’s full biological discovery potential

David Rach, Ph.D.
UMGCCC Flow Cytometry Shared Resource
Baltimore, MD

October 23, 2026

Slides

CC BY SA

Spectral Flow Cytometry is amazing!!!

- Capable of resolving remarkably similar fluorophores → **increased number of markers per panel** compared to conventional flow cytometry (CFC)
- Allows for rapid acquisition of millions of events in a few minutes → typically **greater number of acquired cells per sample** compared to mass cytometry (MC)
- Spectral Flow Cytometry (SFC) datasets have an unique combination of **width and depth** compared to preceding technologies



CC BY SA

Spectral Flow Cytometry is amazing!!!



- Able to efficiently acquire greater number of cells → more thorough profiling of **rare cell subsets**
- Build one panel, evaluate multiple cell types → invaluable for studies with **limited biospecimen**
- Expand our knowledge about immune system heterogeneity

CC BY SA

Trade-off: Increasing analytical complexity

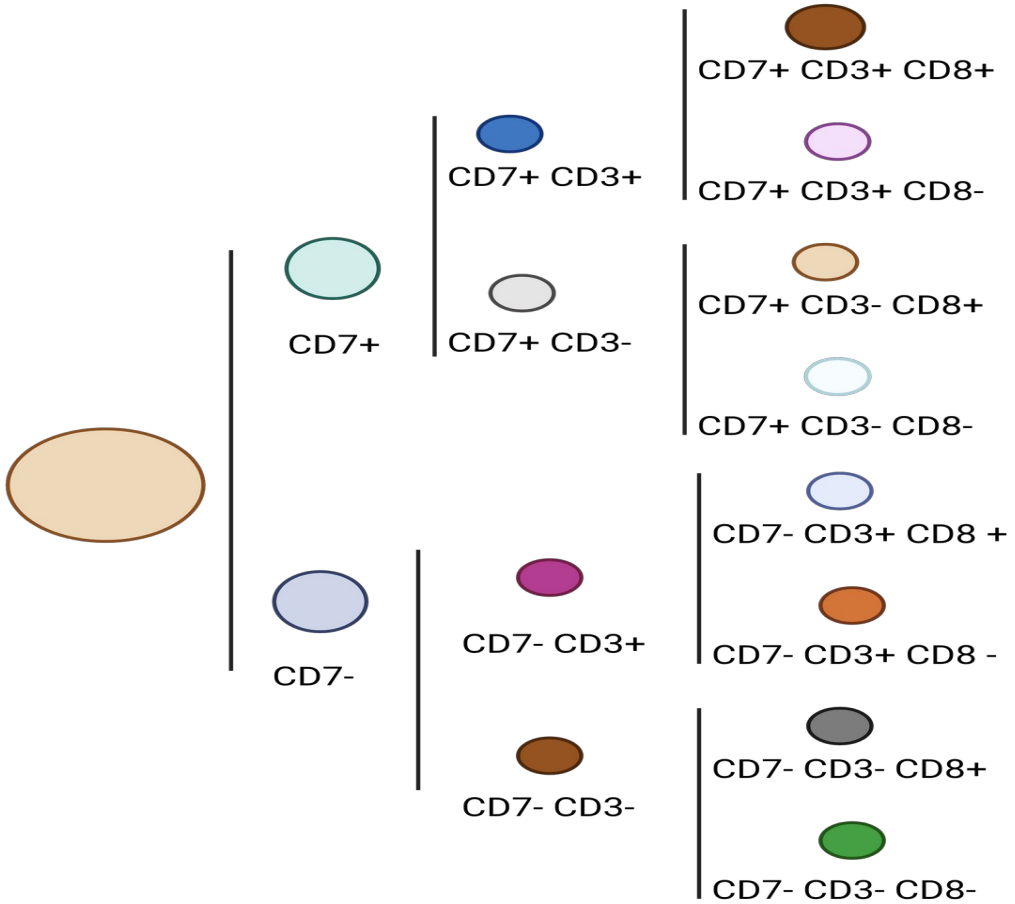
- If you are only interested in a **single population** of cells → analysis process remains <u>essentially unchanged</u> (manual gating with a few more gates)

- If your goal is to characterize **all cell populations** that may be present →
- the breadth and depth of the datasets complicate the analysis



CC BY SA

Complex Datasets: Combinatorially Exponential Clusters

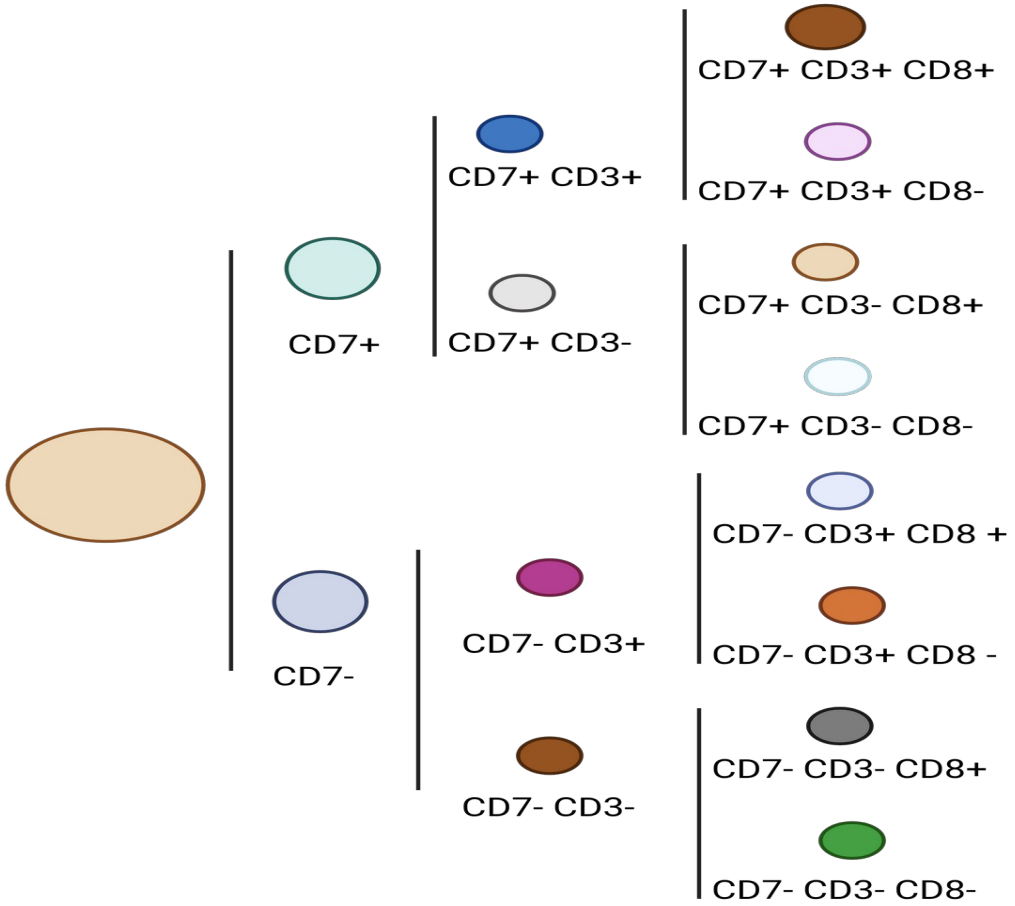


- If for a given marker in your panel, you **split** the cells into **positive** and **negative** populations
- And this splitting process was **repeated for every marker** in your panel
- And at the end, we **grouped cells** that only **expressed the same markers**

- "Groups" per panel = $2^{\text{Number of Fluorophores}}$

CC BY SA

Complex Datasets: Combinatorially Exponential Clusters



- If for a given marker in your panel, you **split** the cells into **positive** and **negative** populations
- And this splitting process was **repeated for every marker** in your panel
- And at the end, we **grouped cells** that only **expressed the same markers**

- **9-color** CFC panel → **512** groups at most
- **40-color** SFC panel → **1.1e+12** groups (**trillion**)
- **60-color** SFC panel → **1.2e+18** groups (**quintillion**)

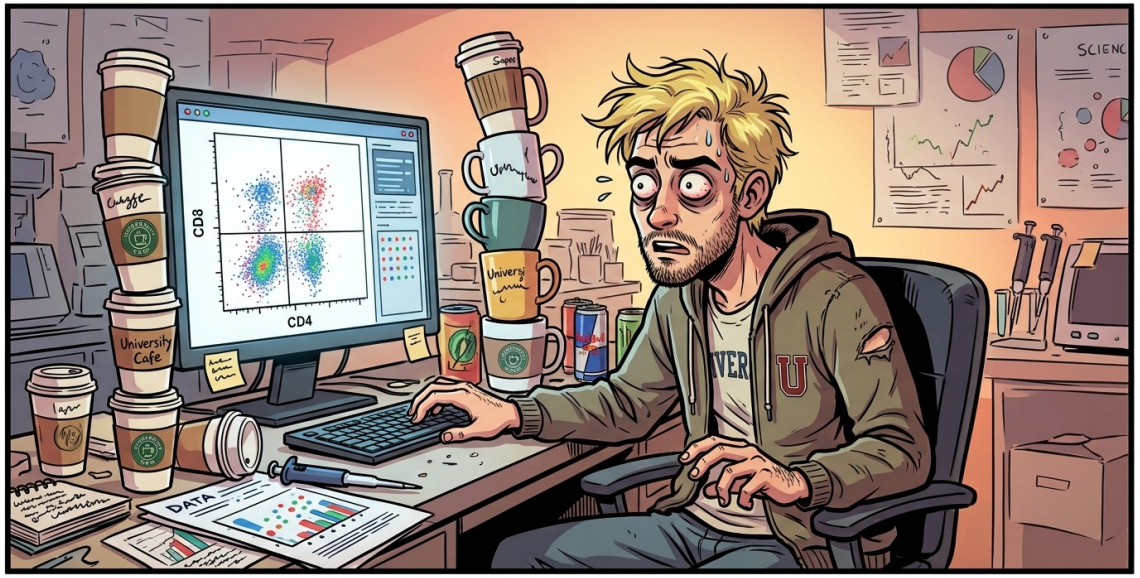
CC BY SA

What even is a quintillion?

If I were to manually check for every combination...

- **60-color** SFC panel → **1.2e+18** groups (**quintillion**)

  - 1 gate, every 10 seconds → 6 groups/minute
  - 60 minutes in an hour → 360 groups/hour
  - Awake 16 hours a day → 5760 groups/day
  - 365 days per year → **2,102,400 groups/year**



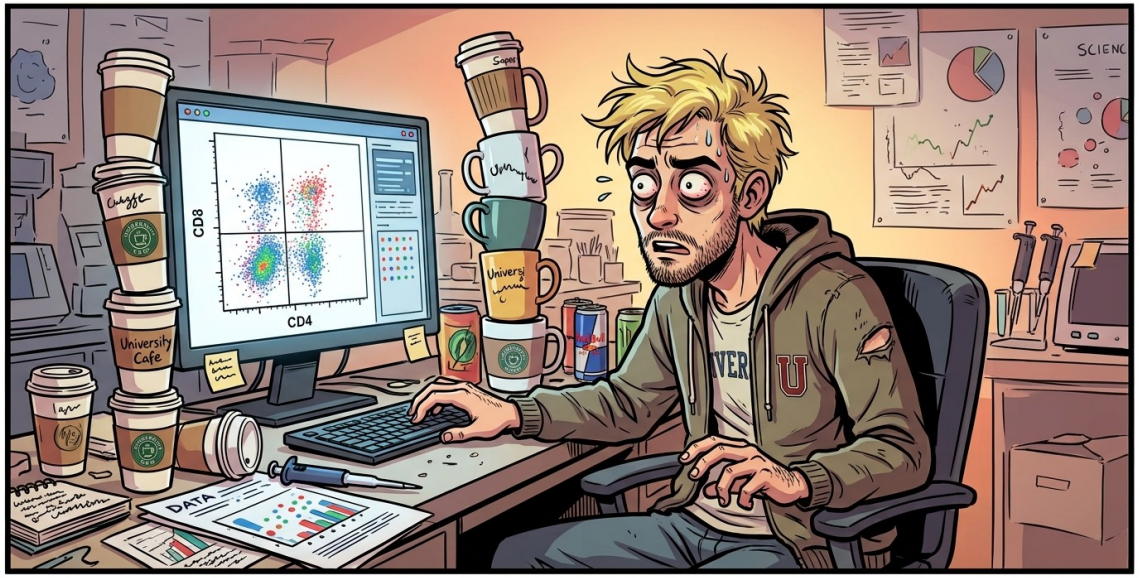
CC BY SA

What even is a quintillion?

If I were to manually check for every combination...

- **60-color** SFC panel → **1.2e+18** groups (**quintillion**)

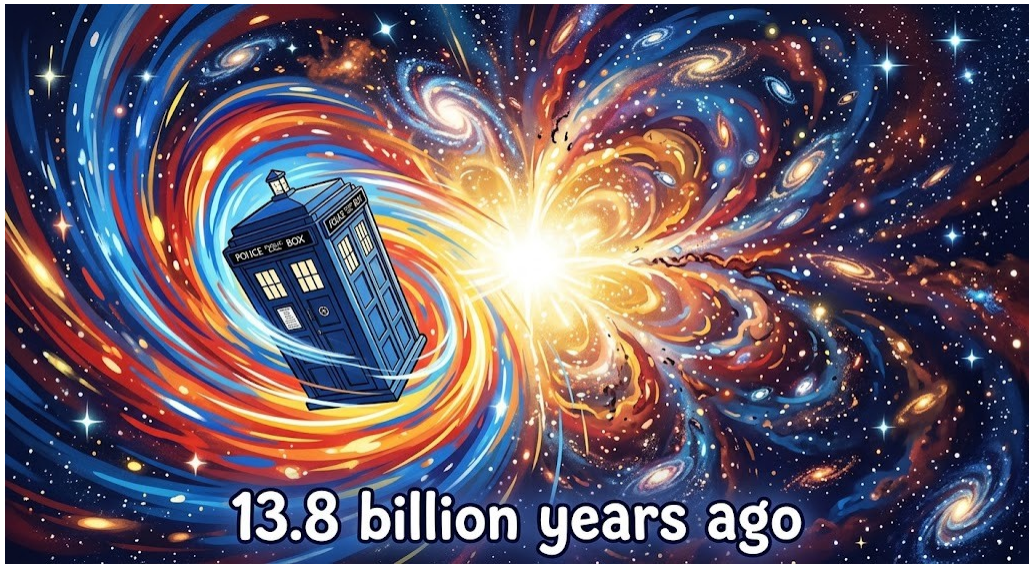
  - 1 gate, every 10 seconds → 6 groups/minute
  - 60 minutes in an hour → 360 groups/hour
  - Awake 16 hours a day → 5760 groups/day
  - 365 days per year → **2,102,400 groups/year**



….Not even close….

CC BY SA

Continuously since the Big Bang

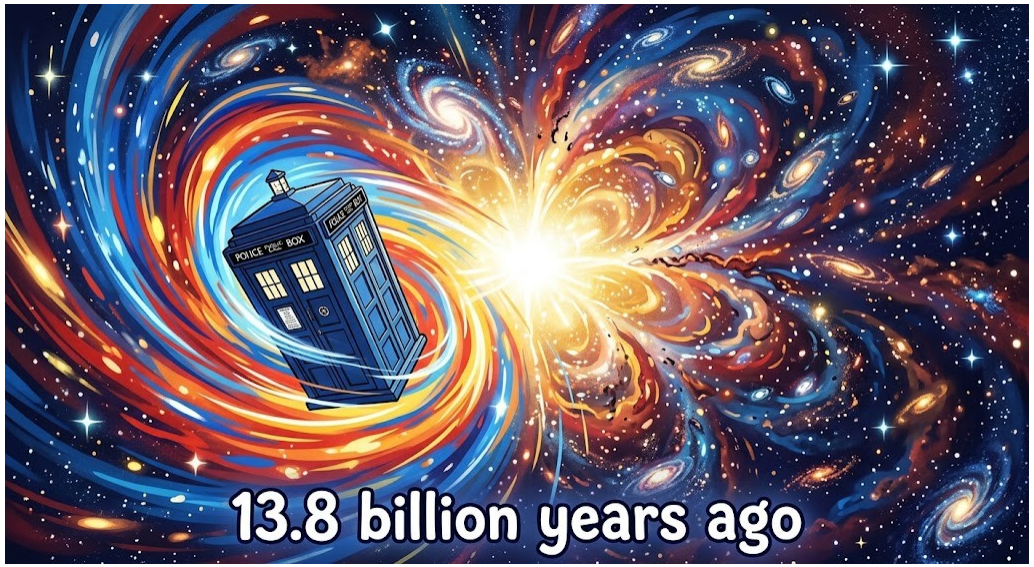


- **60-color** SFC panel → **1.2e+18** groups (**quintillion**)

  - 365 days per year → **2,102,400 groups/year**
  - Invent a time-machine, become immortal
  - 13.8 billion years → **2.9e+16 groups/reality**

CC BY SA

Continuously since the Big Bang



- **60-color** SFC panel → **1.2e+18** groups (**quintillion**)

  - 365 days per year → **2,102,400 groups/year**
  - Invent a time-machine, become immortal
  - 13.8 billion years → **2.9e+16 groups/reality**

….Still not enough….

CC BY SA

Continuously since the Big Bang



- **60-color** SFC panel → **1.2e+18** groups (**quintillion**)

  - 365 days per year → **2,102,400 groups/year**
  - Invent a time-machine, become immortal
  - 13.8 billion years → **2.9e+16 groups/reality**

  - So we need ~ <u>50 immortal researchers working continuously since the Big Bang</u> just to check all group combinations for a **<u>single 60-fluorophore panel .fcs file</u>**

CC BY SA

Continuously since the Big Bang



- **60-color** SFC panel → **1.2e+18** groups (**quintillion**)

  - 365 days per year → **2,102,400 groups/year**
  - Invent a time-machine, become immortal
  - 13.8 billion years → **2.9e+16 groups/reality**

  - So we need ~ <u>50 immortal researchers working continuously since the Big Bang</u> just to check all group combinations for a **<u>single 60-fluorophore panel .fcs file</u>**

1 file down, 99 more to go!

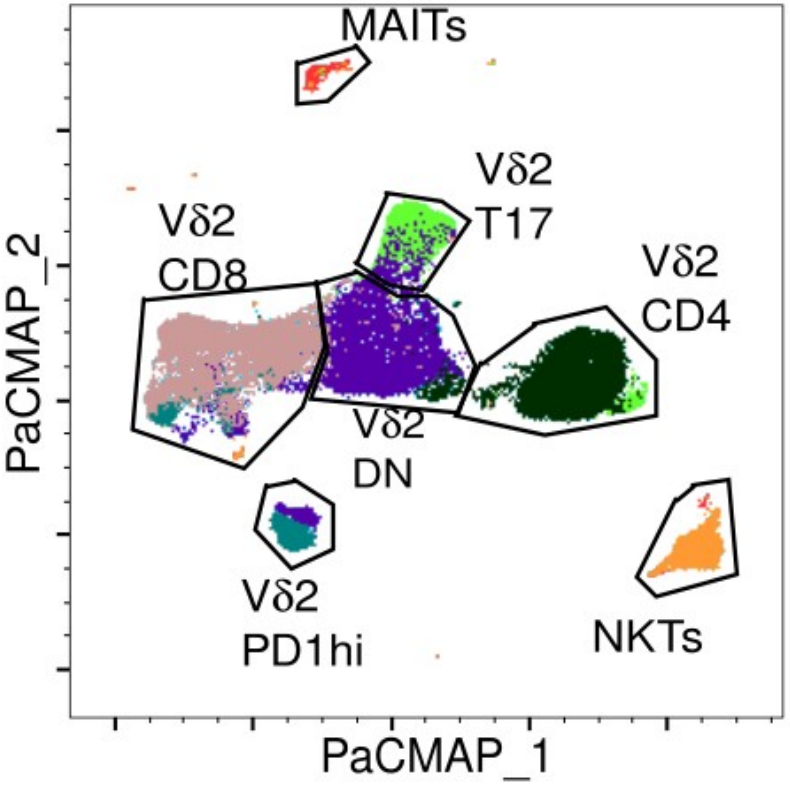
CC BY SA

Alternative: Unsupervised Analysis (Algorithms)

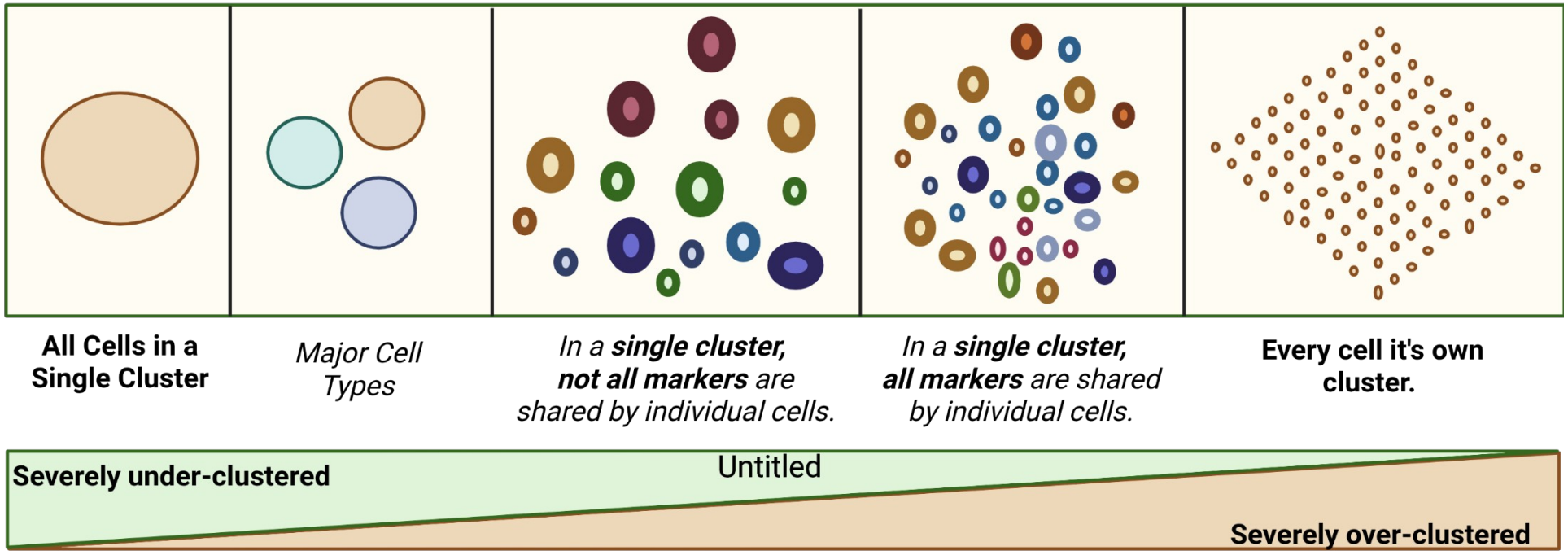
- Consequently, if the goal is to characterize all cell populations present, some form of **unsupervised (algorithm) or semi-supervised** analytical approaches will be needed

- **Clustering** → FlowSOM, Phenograph, Xshift, etc.
- **Dimensionality visualization** → opt-SNE, UMAP, PaCMAP, Phate, etc.

- **Automated Gating** → openCyto, flowMagic, ElasticGate, etc.

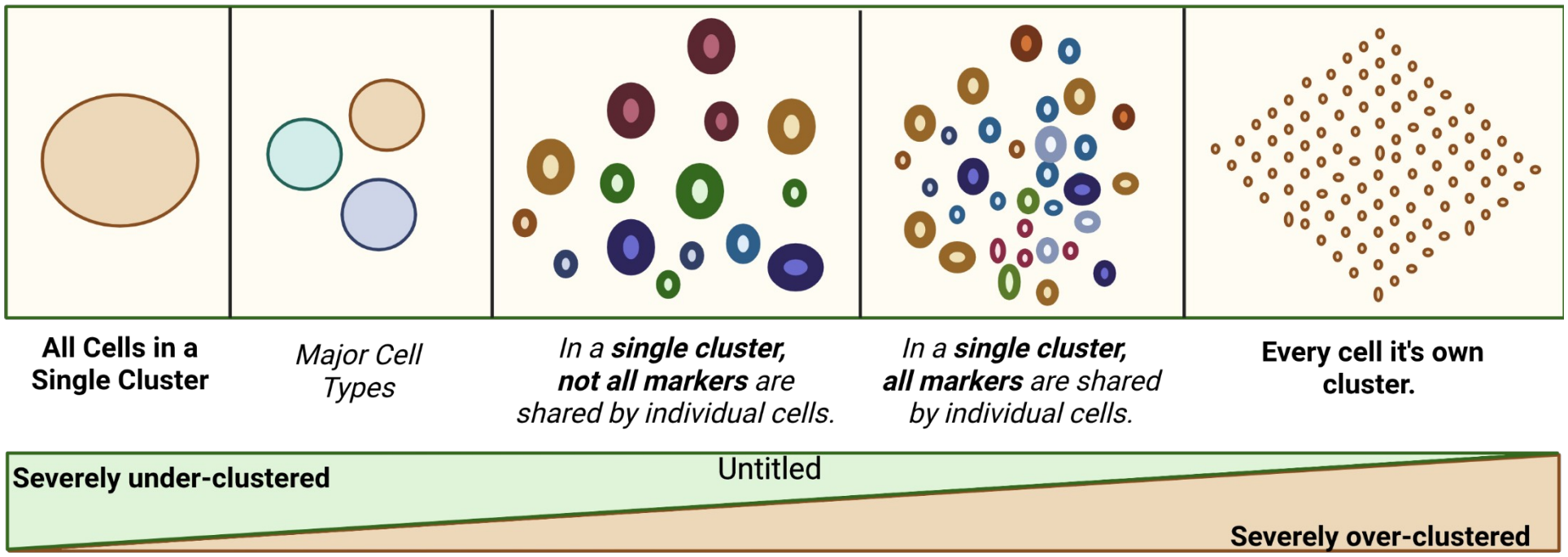


The challenge for unsupervised approaches



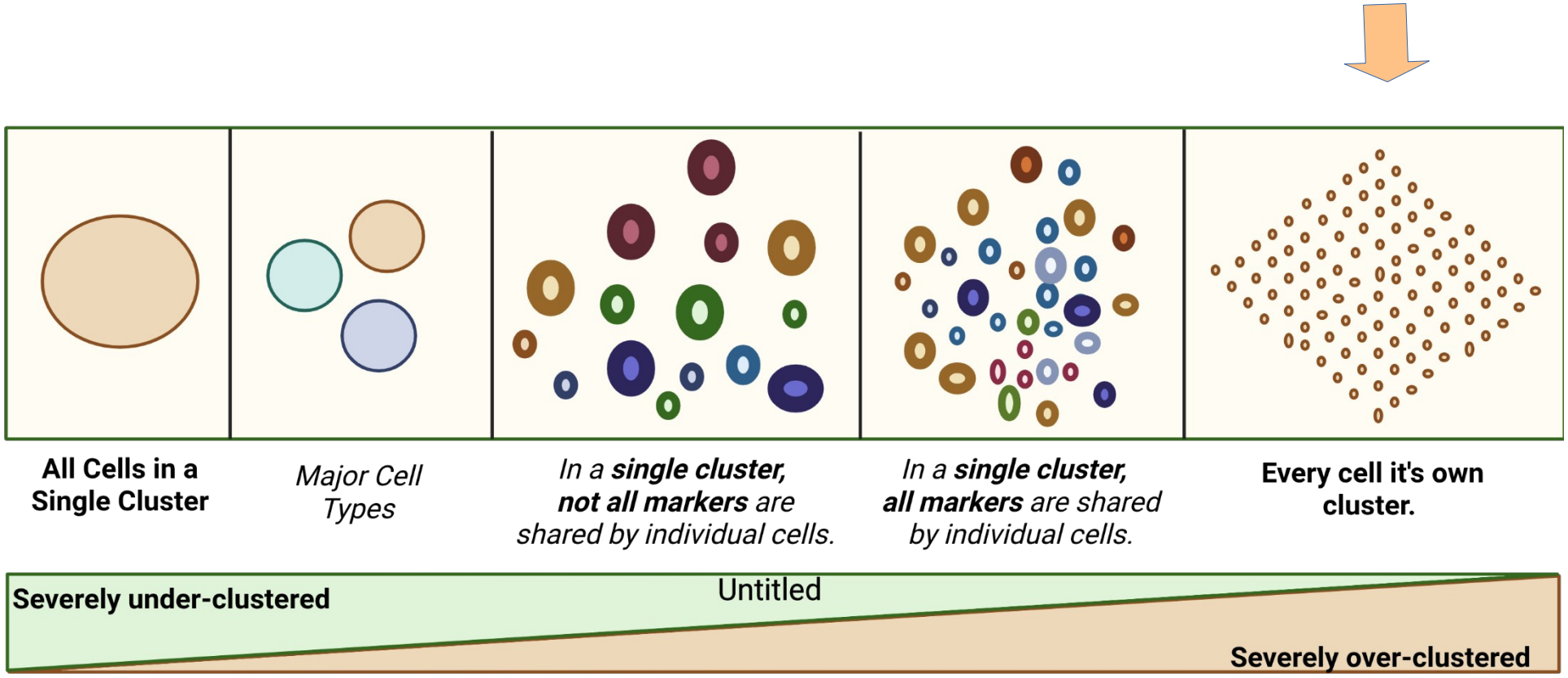
CC BY SA

The challenge for unsupervised approaches



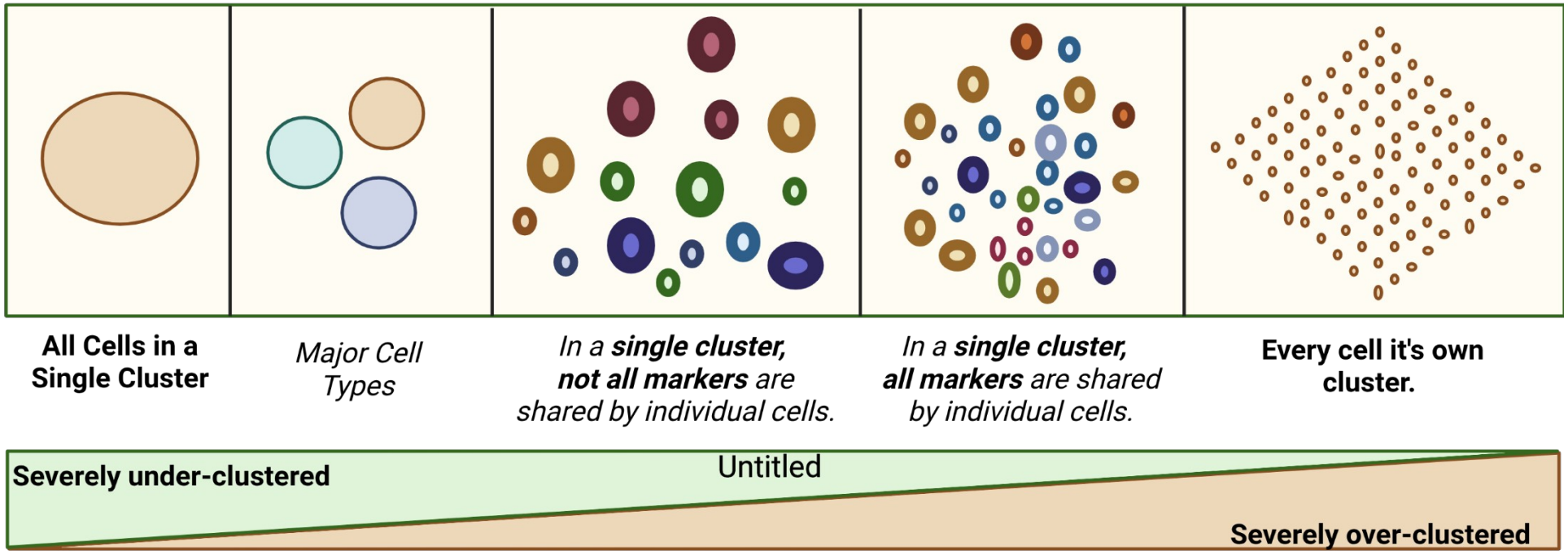
CC BY SA

The challenge for unsupervised approaches



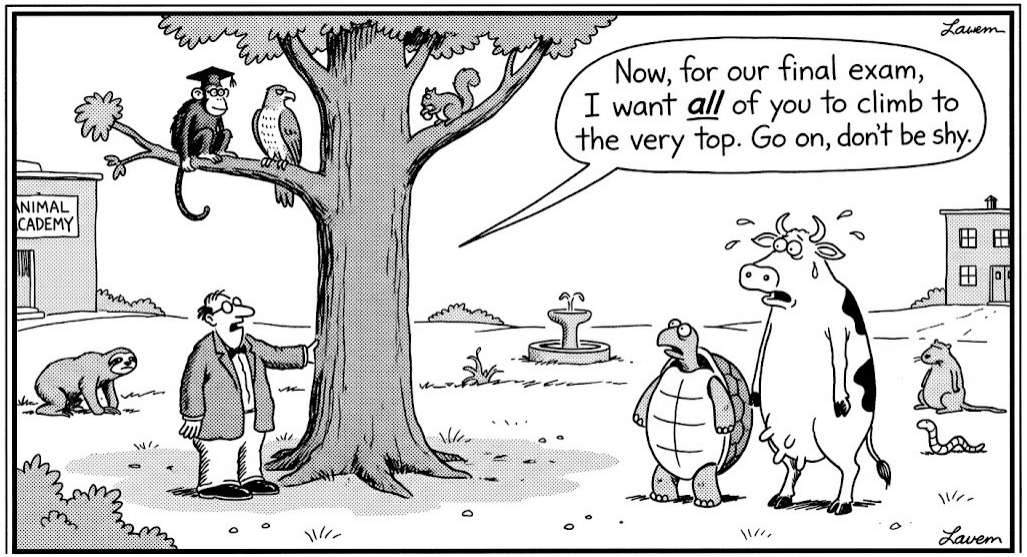
CC BY SA

The challenge for unsupervised approaches



CC BY SA

Adapted algorithmic tools struggle with number of events



- Many of these software tools were originally implemented as R or Python packages, for use in analysis of **mass cytometry** or **single-cell RNA-seq datasets**
- Mass cytometry → similar number of markers, typically **fewer events**, no unmixing variation due to autofluorescence
- Single-cell RNA-seq → substantially more markers, typically way **fewer events.** No equivalent "negative" gate events for not expressed transcripts

- **Down-sampling becomes often necessary**

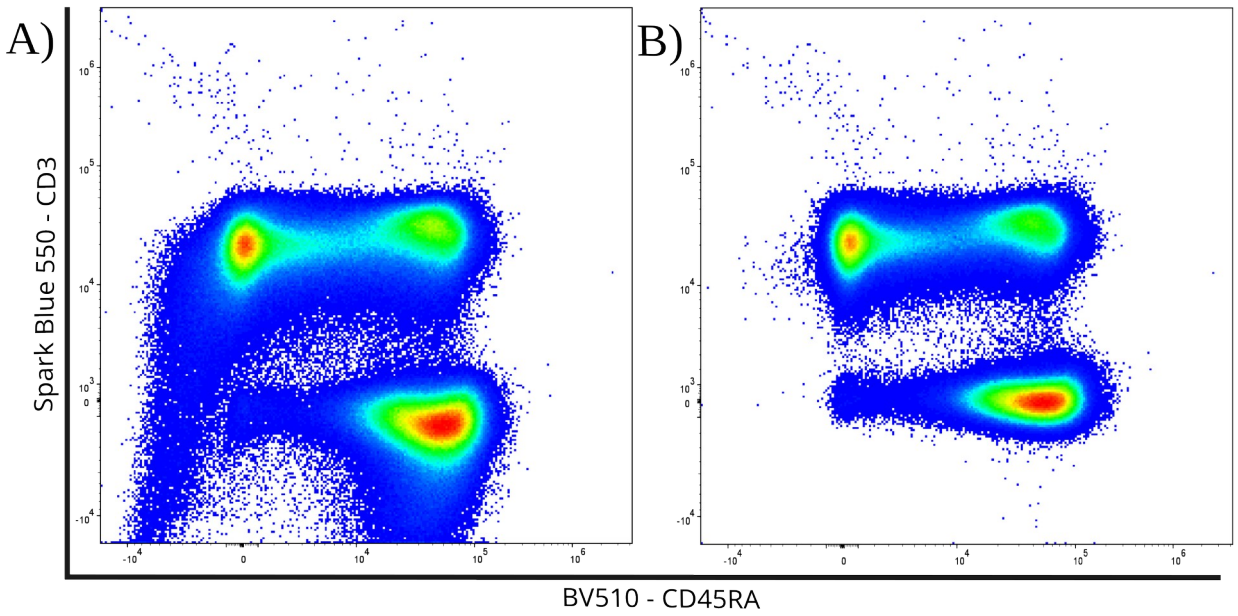
CC BY SA

Adapted algorithmic tools struggle with unmixing errors

- Adapted software packages heavily rely on having a consistent **median fluorescent intensity (MFI)** to cluster cells based on positive or negative marker expression

- Existing **batch correction** / normalization / scaling tools target instrumental or **experimental** level variation for correction

- **Unmixing errors** can seriously shift the MFI ranges for “negative” and “positive” cells for **individual** acquired specimens



CC BY SA

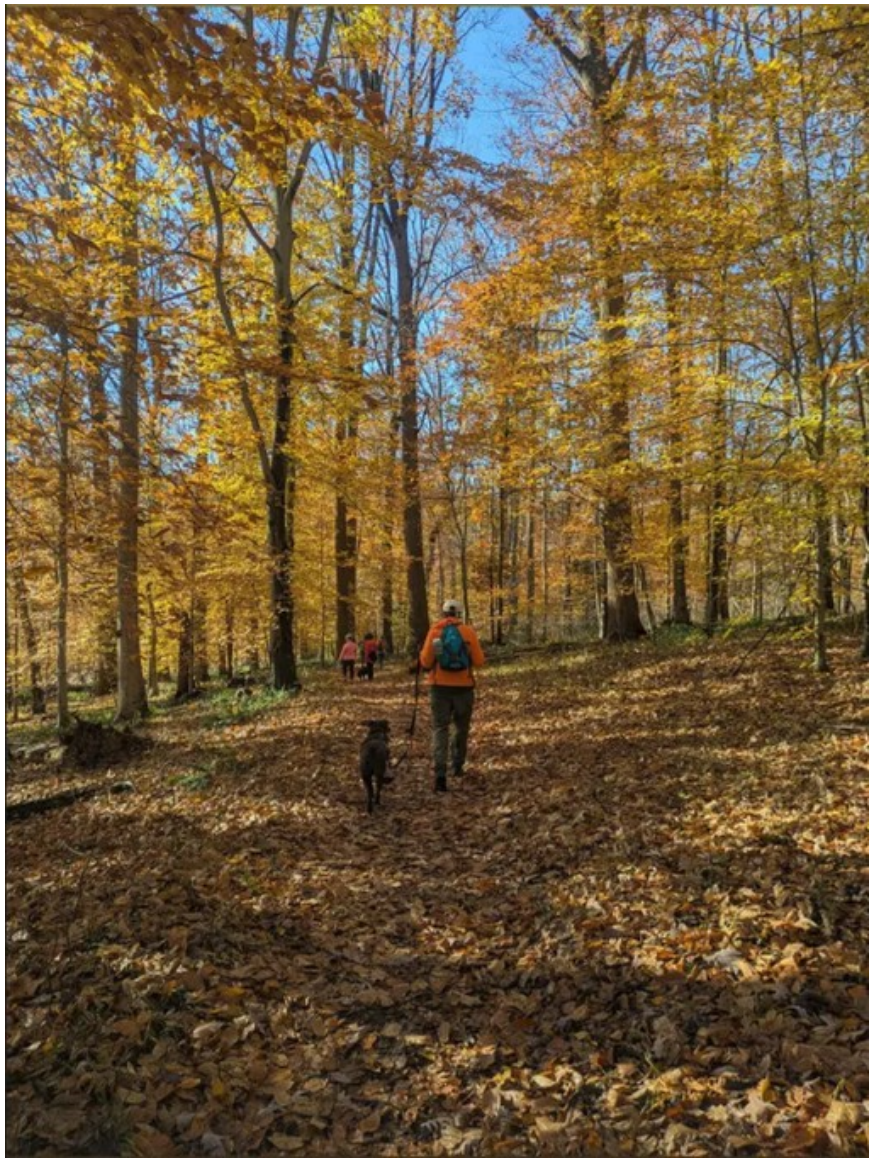
Full biological discovery potential often out of reach



- Most flow cytometrist don’t code
- You likely lost your bioinformatician while describing manual gating
- The statisticians were never consulted
- For **most** spectral flow cytometry datasets, only a **handful of cell populations** end up being analyzed by any method

CC BY SA

Personal Goal: No cell left behind



Repeated 3 AM thoughts

- ❑ What makes spectral flow cytometry data amazing is both the depth and the breadth

- ❑ How to fully profile the entirety of the acquired dataset, not just a single cell population of interest

CC BY SA

Building open-source tool-sets to address SFC's quirks



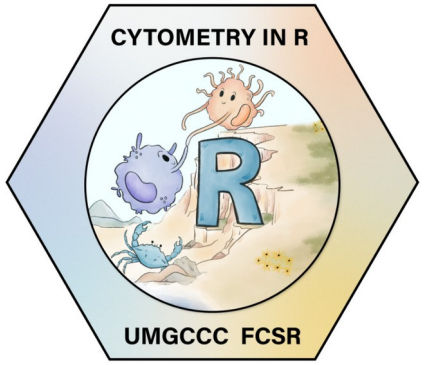
Spectral Signature Characterization and Isolation

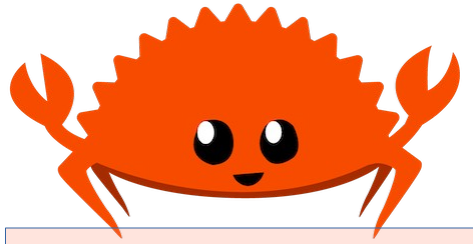


Incorporation of manually-annotated gating information



Automated Instrument Quality-Control Dashboards

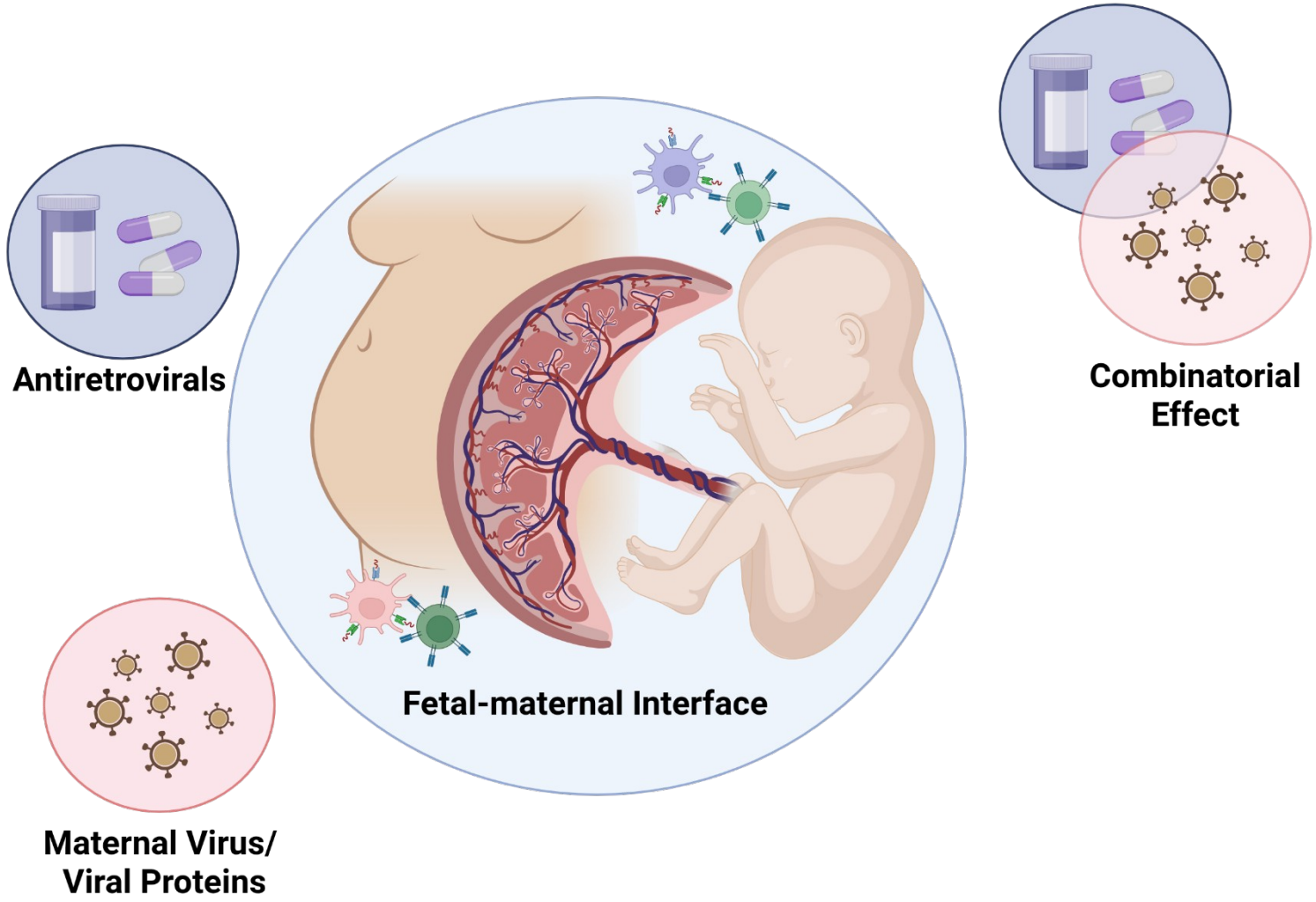




Rust

CC BY SA

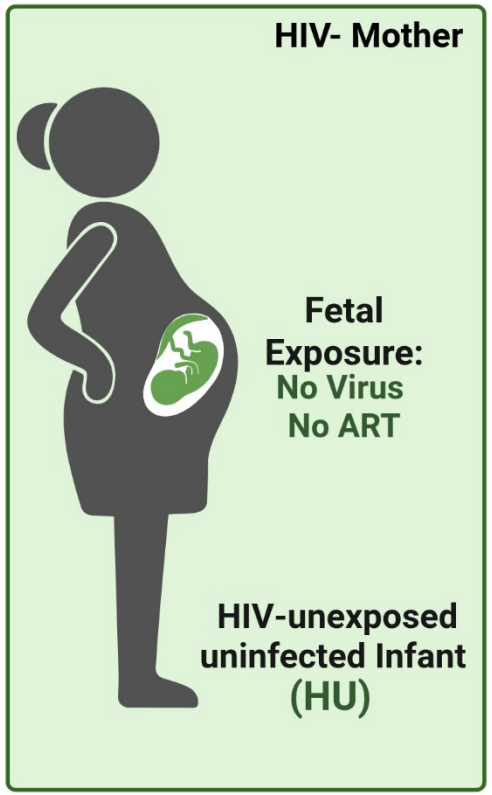
My PhD's focus: HIV-exposed Uninfected Infants

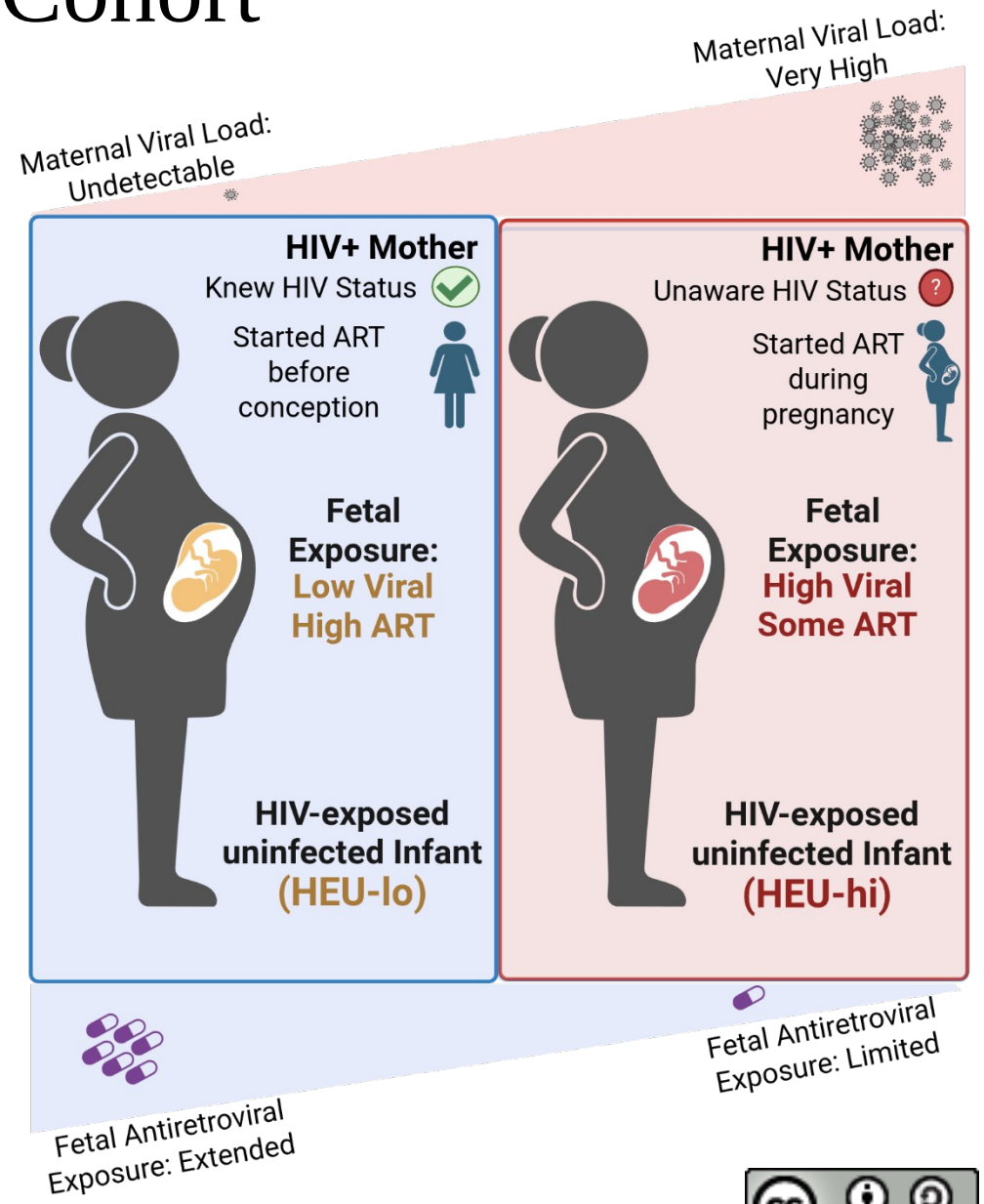


CC BY SA

Our Malawian Cohort



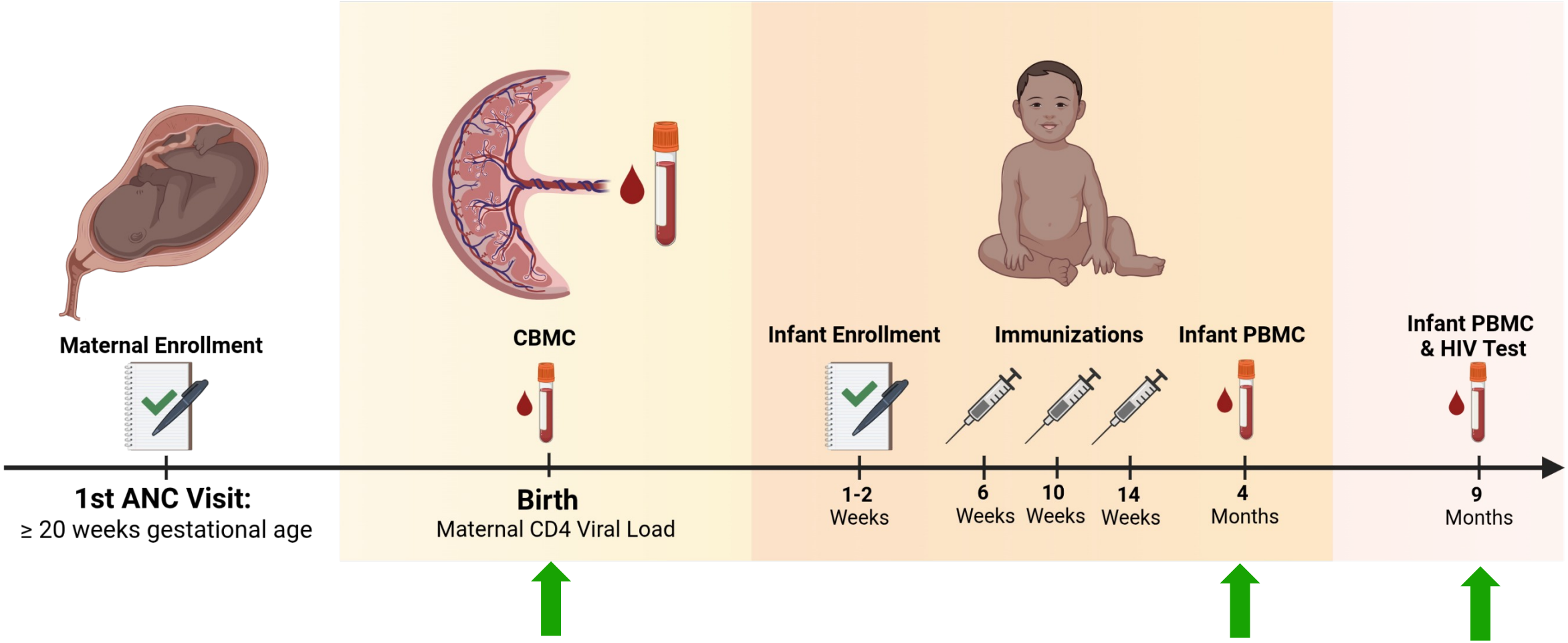




UNIVERSITY of MARYLAND
SCHOOL OF MEDICINE

CC BY SA

Study Timeline



UNIVERSITY of MARYLAND SCHOOL OF MEDICINE

CC BY SA

32-Fluorophore Spectral Flow Cytometry Panel

UV1	Spark UV 387		
UV2	BUV395	CD62L	(SK11)
UV6	cFluor UV440		
UV7	BUV496		
UV9	BUV563	CD69	(FN50)
UV10	BUV615	CCR4	(1G1)
UV11	BUV661	Vδ2	(B6)
UV14	BUV737	CD38	(HB7)
UV16	BUV805	CD4	(SK3)
V1	BV421/		
V2	Spark Violet 423		
V3	Pacific Blue	CD14	(M5E2)
V3	Pacific Blue	CD19	(SJ25C)
V4	SYTOX Blue		
V5	BV480	CD161	(HP-3G10)
V6	SBV515		
V7	BV510		
V8	BV570	CD16	(3G8)
V9	Odot 585		
V10	BV605	CD45RA	(HI100)
V11	BV650	CD8	(RPA-T8)
V13	BV711	Va7.2	(3C10)
V14	BV750	IFNγ	(B27)
V15	BV786	CCR6	(11A9)
V16	LD NIR 786		
B2	Alexa Fluor 488	FoxP3	(236A/E7)
B3	Spark Blue 550	CD3	(SK7)
B4	Spark Blue 574	CD45	(2D1)
B6	RB613	PD1	(EH12.1)
B7	NovaFluorBlue660		
B8	SBB675		
B9	NovaFluorBlue690		
B10	RB705	CD26	(M-A261)
B12	RB744/		
B13	NovaFluorBlue760		
B14	RB780	CXCR5	(RF8B2)
YG1	PE	ICOS	(ISA-3)
YG2	Spark YG 593		
YG3	PE-Dazzle 594	TNFα	(MAB11)
YG4	PE-Fire 640/		
YG5	PE-Cy5	CXCR3	(G025H7)
YG6	NovaFluorYellow690		
YG7	PE-Fire 700	CD127	(A019D5)
YG8	PE-Fire 744	CD25	(M-A251)
YG9	PE-Vio 770	HLA-DR	(REA805)
YG10	PE-Fire810		
R1	APC	CD39	(A1)
R2	Alexa Fluor 647	IL2	(MQ1-
R3	cFluor R685		
R4	APC-R700	CD107a	(H4A3)
R5	NovaFluorRed725		
R6	Zombie NIR	L/D	
R7	APC-Fire 750	CD27	(O323)
R8	APC-Fire 810	CCR7	(G043H7)

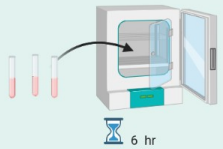
Thaw & Rest



1) Thaw



2) Count

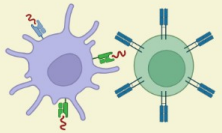


3) Rest

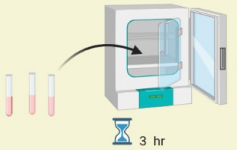
Antigen Incubation



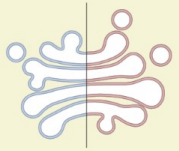
4) Aliquot by condition



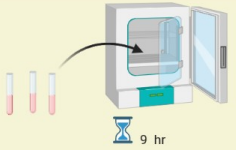
5) Add antigens (PPD, SEB, Ctrl)



6) Incubate



7) Add Golgi Transport Inhibitors



8) Incubate

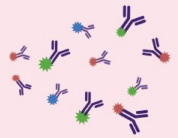
Flow Cytometry Staining



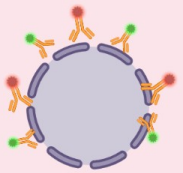
9) Pellet



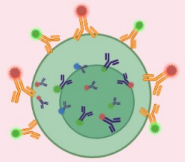
10) Viability stain



11) Surface antibodies



12) Fix & permeabilize



13) Intracellular antibodies



14) Resuspend in 0.4% PFA



15) Refrigerate at 4*C

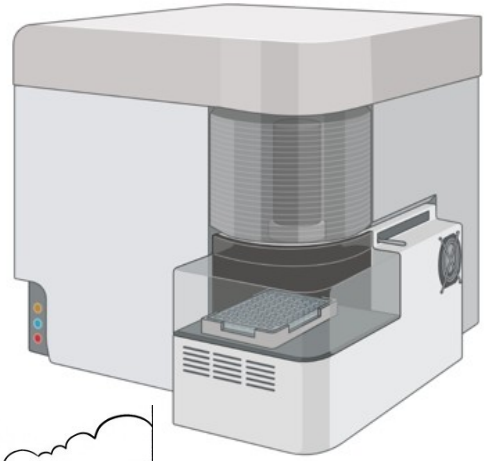


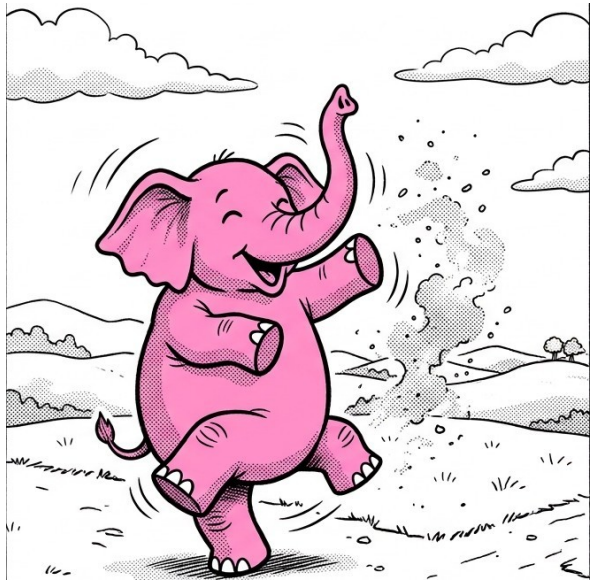
16) Acquisition Cytek Aurora 5L

UNIVERSITY of MARYLAND
SCHOOL OF MEDICINE

CC BY SA

To unlock the comprehensive biological discovery potential, we need to address these pink elephants



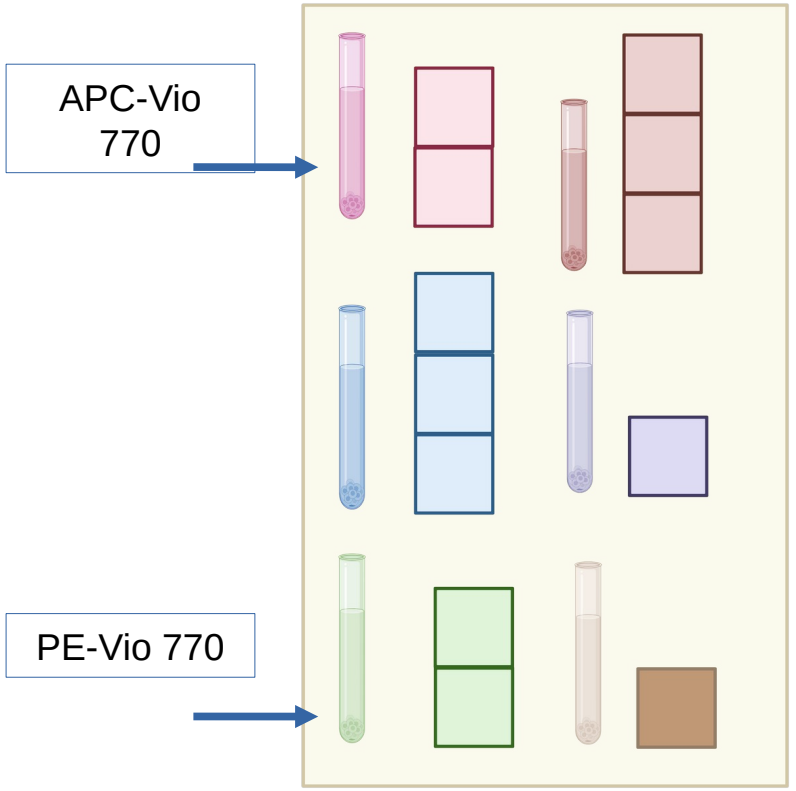


- Quality of the unmixed full-stained data is linked to the quality of our single-color controls (and our ability to provide appropriate autofluorescence controls).

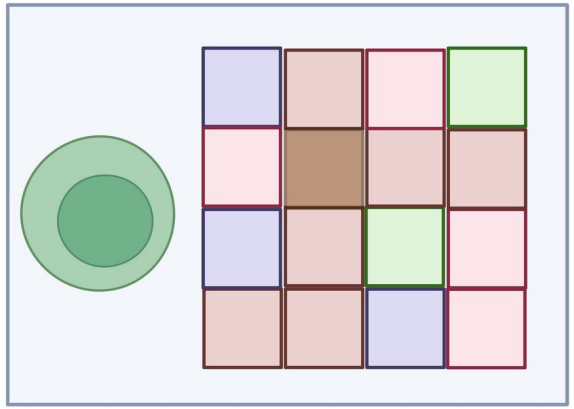
- If we are acquiring 3 million cells to detect 100 rare cells, how do we thoroughly explore the rest of the sample to avoid wasting useful data?

CC BY SA

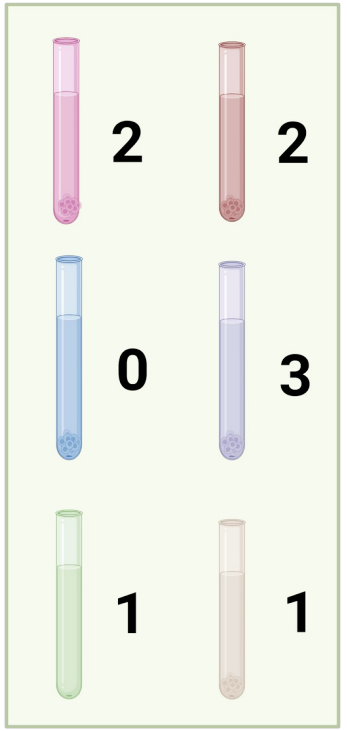
Spectral Unmixing



Unmixing Controls (Single Color and Unstained)



Single cell from full-stained sample



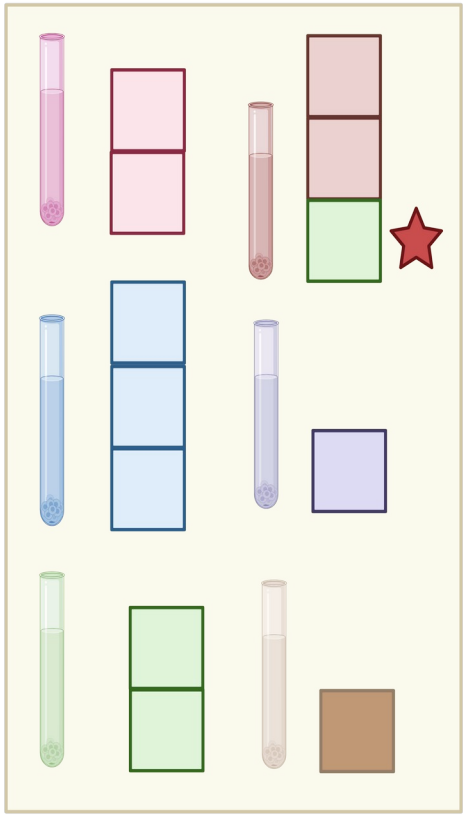
What fluorophores are present, and roughly how much

CC BY SA

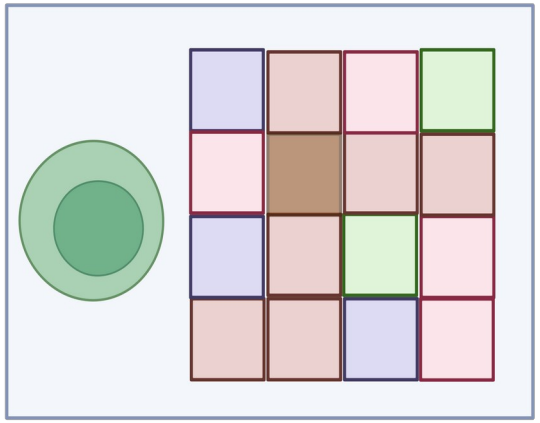
Mismatched References lead to errors in attribution

APC-Vio 770

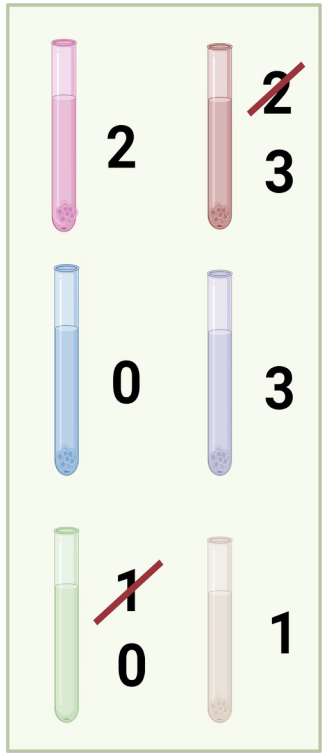
PE-Vio 770



Unmixing Controls (Single Color and Unstained)



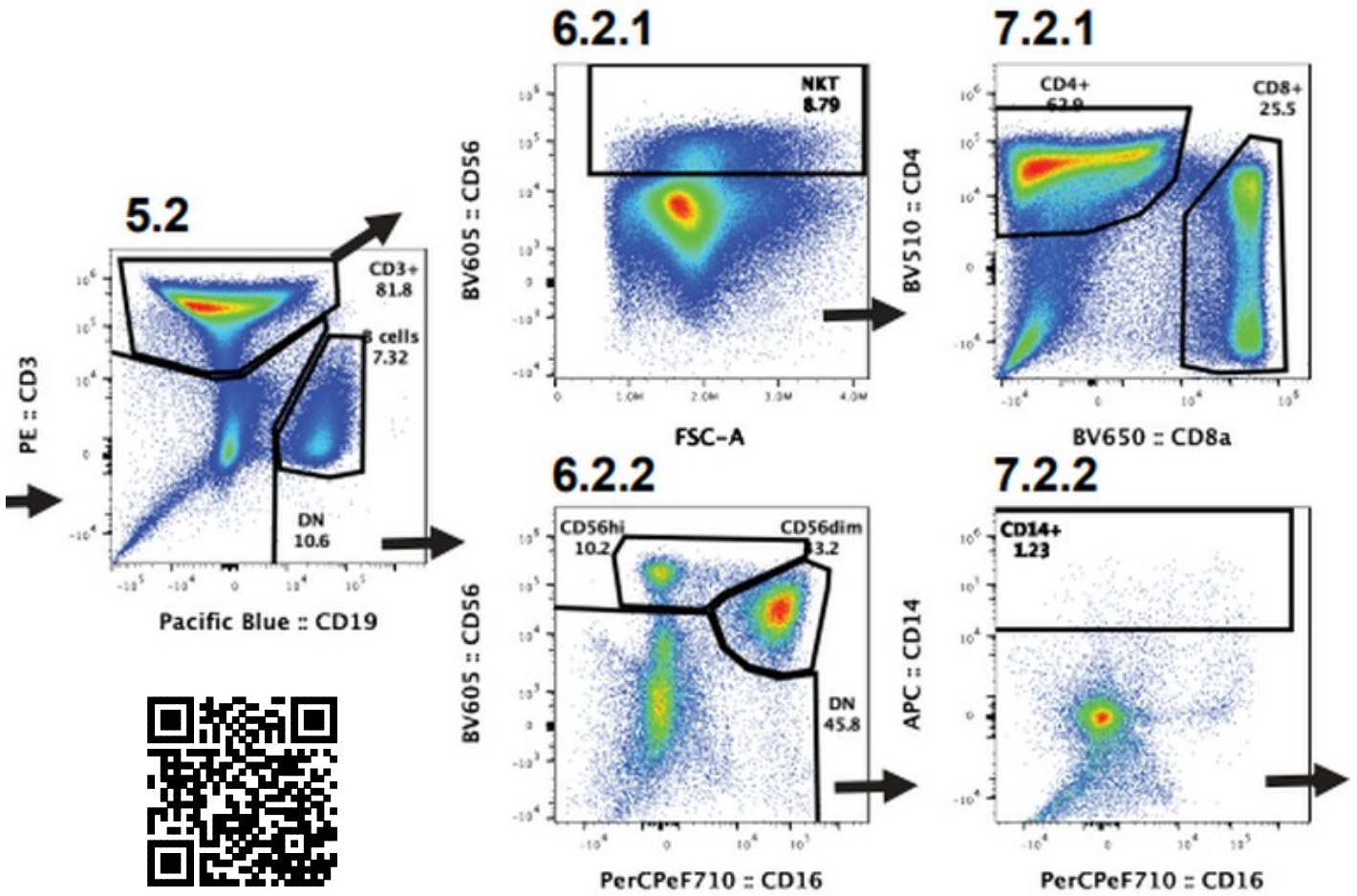
Single cell from full-stained sample



What fluorophores are present, and roughly how much

CC BY SA

“We discovered a novel dendritic cell cluster”



- Despite the critical importance unmixing controls play, few tools permit us to determine quality of the individual control before unmixing
- Existing quality control algorithms (PeacoQC, FlowAI, etc.) primarily focus on detecting laser and fluidic issues in already acquired .fcs files.
- Unmixing errors can be introduced by tandem degradational, additional autofluorescences, undiagnosed instrumental issues, etc.

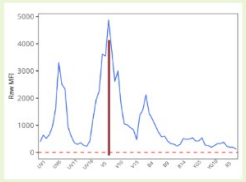
University of Maryland School of Medicine

CC BY SA

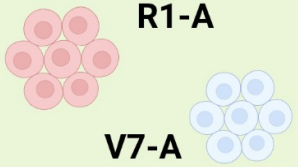
Luciernaga



Code



Identify detector with brightest MFI for each cell



Group cells with shared peak detectors



Subtract Out Autofluorescence

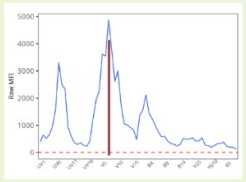
University of Maryland School of Medicine

CC BY SA

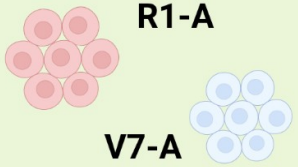
Luciernaga



Code



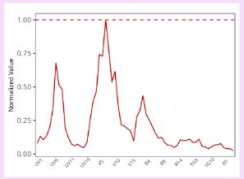
Identify detector with brightest MFI for each cell



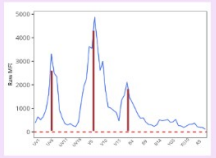
Group cells with shared peak detectors



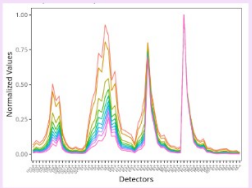
Subtract Out Autofluorescence



Normalize individual cell fluorescent signatures



Identify peaks using local maxima



Group cells with shared peaks & heights

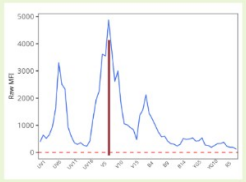
University of Maryland School of Medicine

CC BY SA

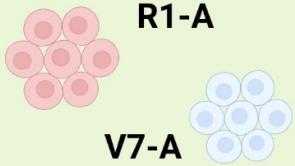
Luciernaga



Code



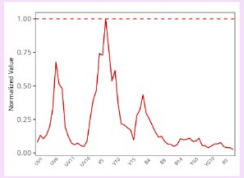
Identify detector with brightest MFI for each cell



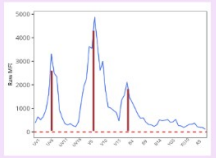
Group cells with shared peak detectors



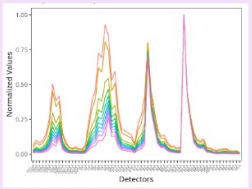
Subtract Out Autofluorescence



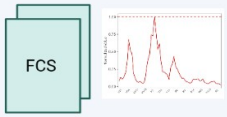
Normalize individual cell fluorescent signatures



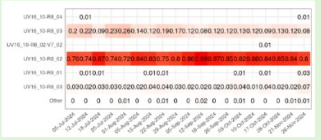
Identify peaks using local maxima



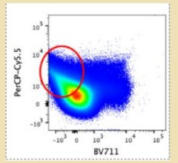
Group cells with shared peaks & heights



Enumerate and export out cells that share signature as .fcs files



Checks for tandem degradation

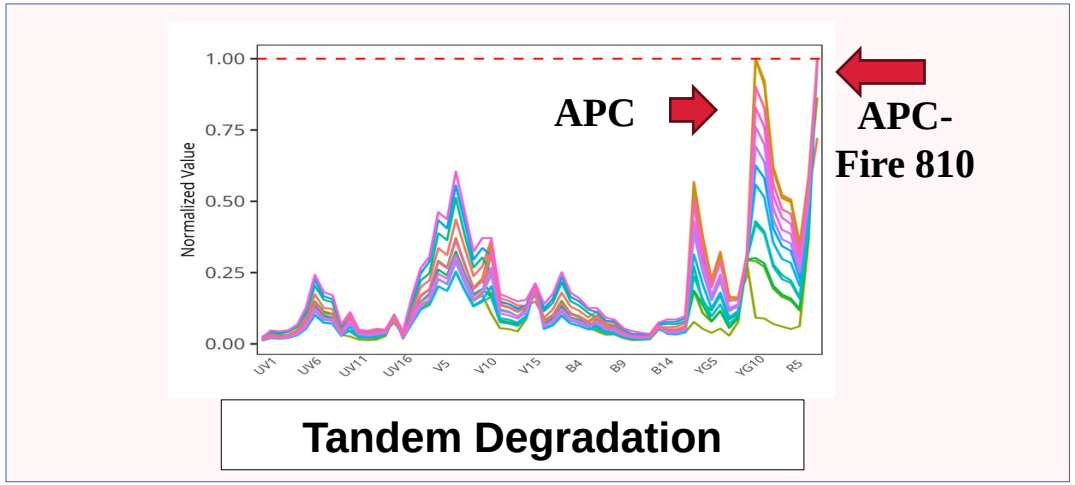


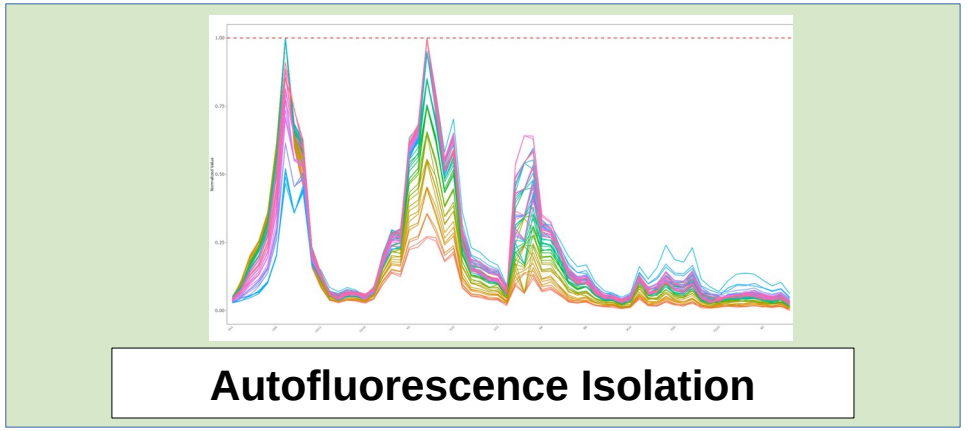
Autofluorescence Isolation

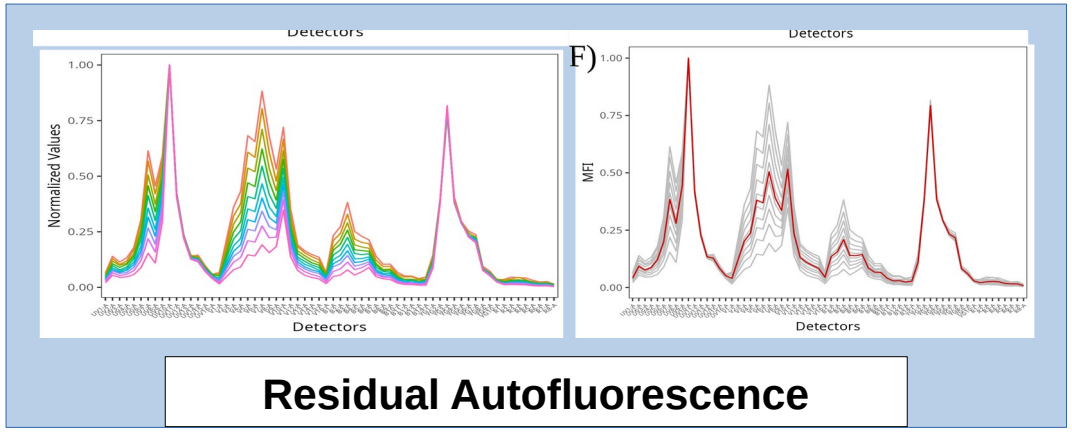
UNIVERSITY of MARYLAND
SCHOOL OF MEDICINE

CC BY SA

Automated tools for screening unmixing controls







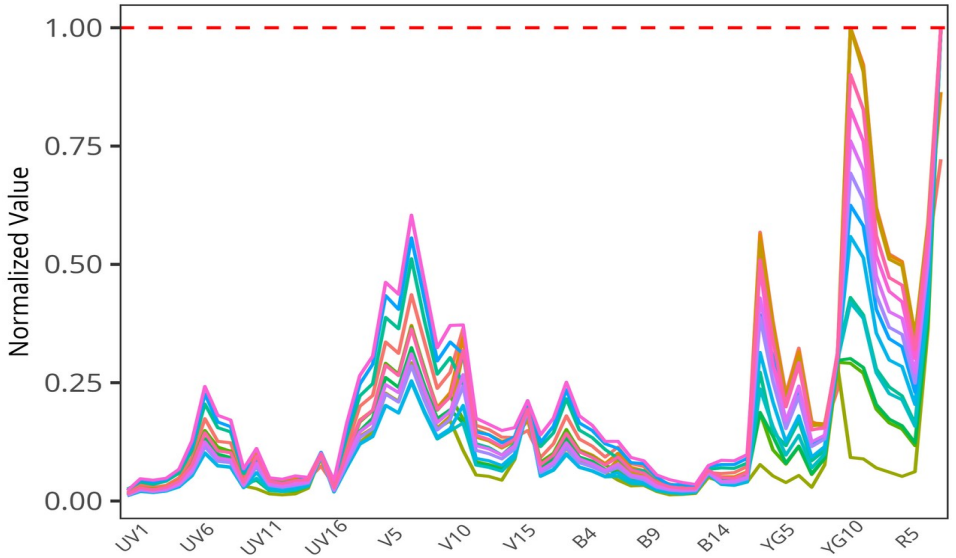
CC BY SA

Identifying degrading fluorophores

Fluorophores in good shape → **one predominant normalized signature** within the unmixing control raw .fcs file

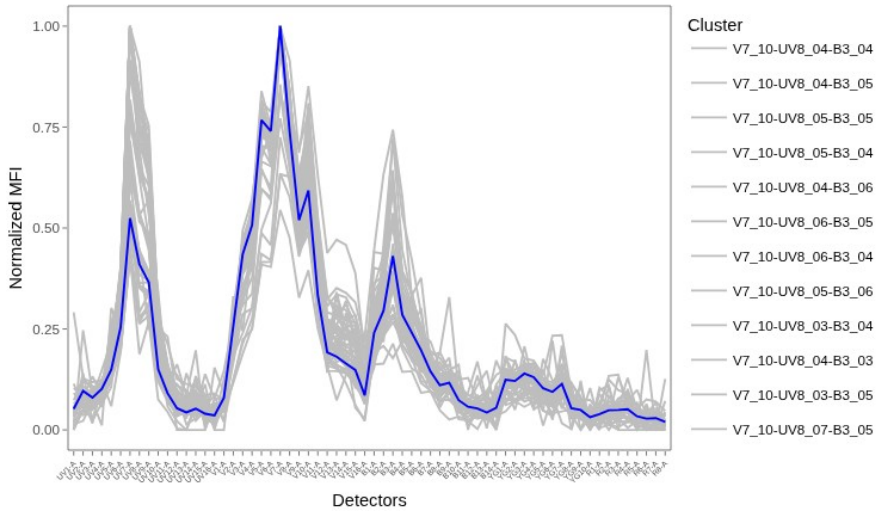
- When the fluorophore degrades, noticeable **increase in variant signatures**, making up substantial proportion of all cells in the unmixing control

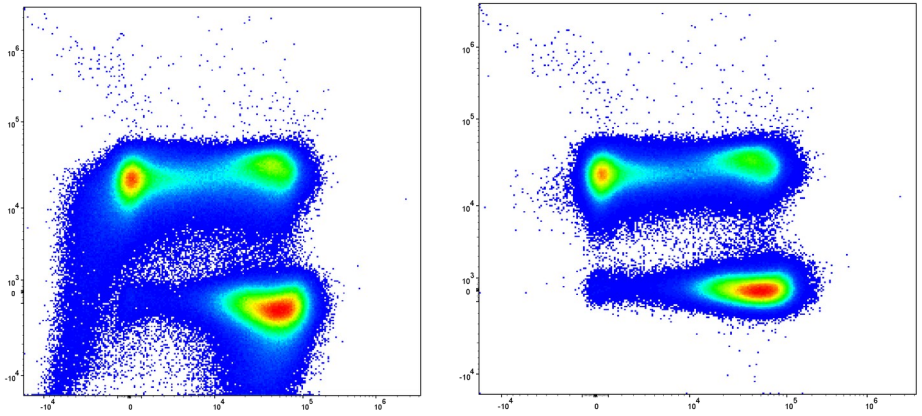
- Automated screening all acquired unmixing controls from campus users during off-hours



CC BY SA

Isolating additional autofluorescence signatures





- This approach also works to isolate variant autofluorescence signatures from unstained controls

- Unlike other approaches, the isolated cells share normalized signatures, only differing in terms of brightness

- Return directly as an .fcs file containing just cells with same signature for use as an autofluorescent tag in SpectroFlo

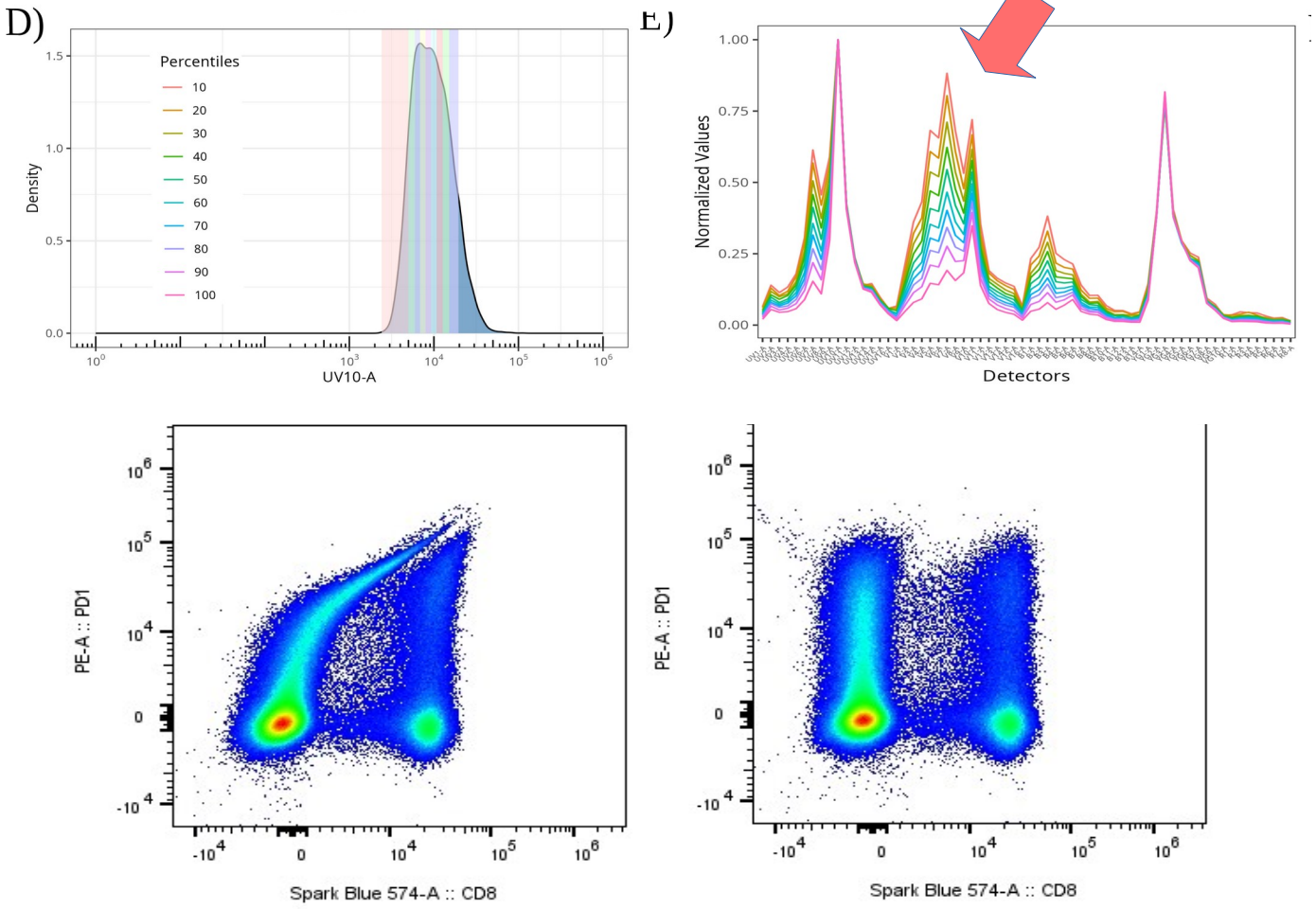
CC BY SA

Residual autofluorescence in dim cell single-color controls

- **“As bright or brighter”**

- Dim antigen-fluorophore combinations acquired as **cell single-color controls** often have residual autofluorescence contribution to their signature

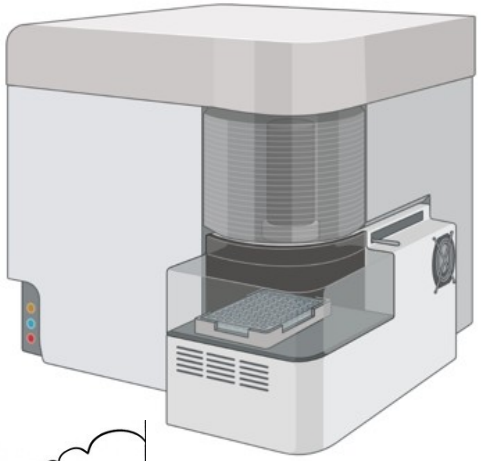
- Identifying and **switching these** dim cell single-color controls to **bead single-color controls** significantly improved unmixing for our panels

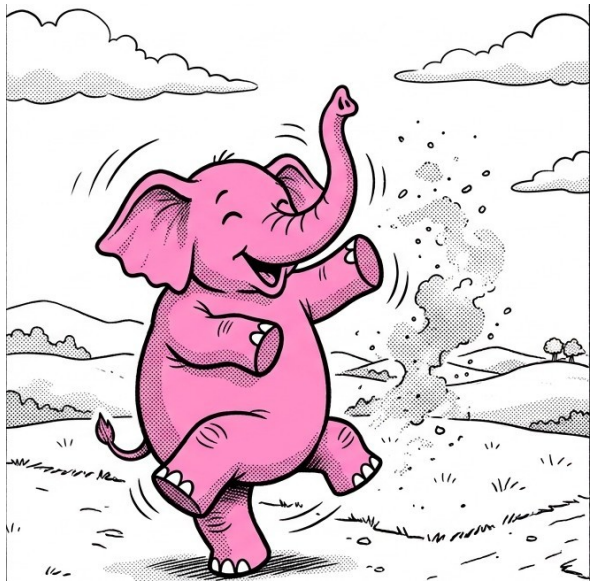


Signature with residual AF

Signature without residual AF

To unlock the comprehensive biological discovery potential, we need to address these pink elephants



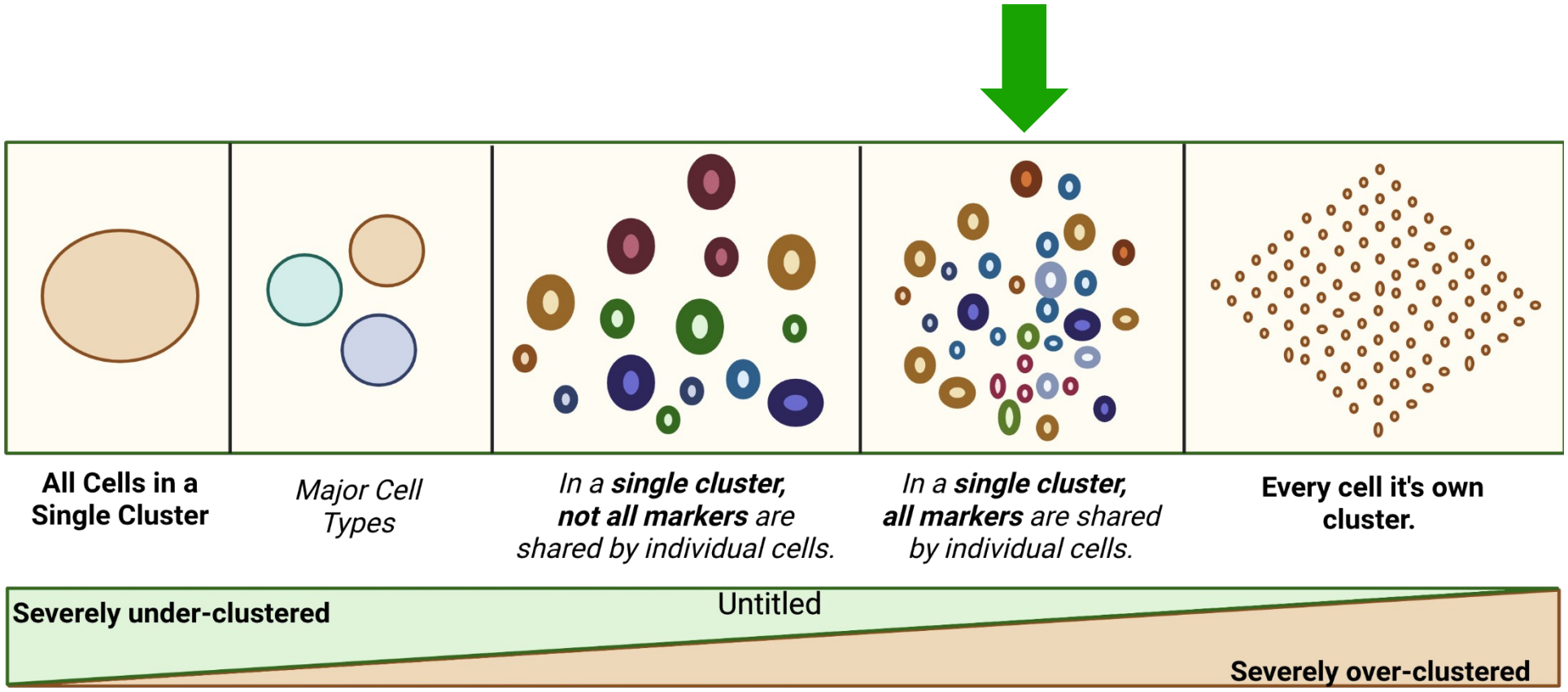


- [x] Quality of the unmixed full-stained data is linked to the quality of our single-color controls (and our ability to provide appropriate autofluorescence controls).

- [ ] If we are acquiring 3 million cells to detect 100 rare cells, how do we thoroughly explore the rest of the sample to avoid wasting useful data?

CC BY SA

Grouping cells that share every marker



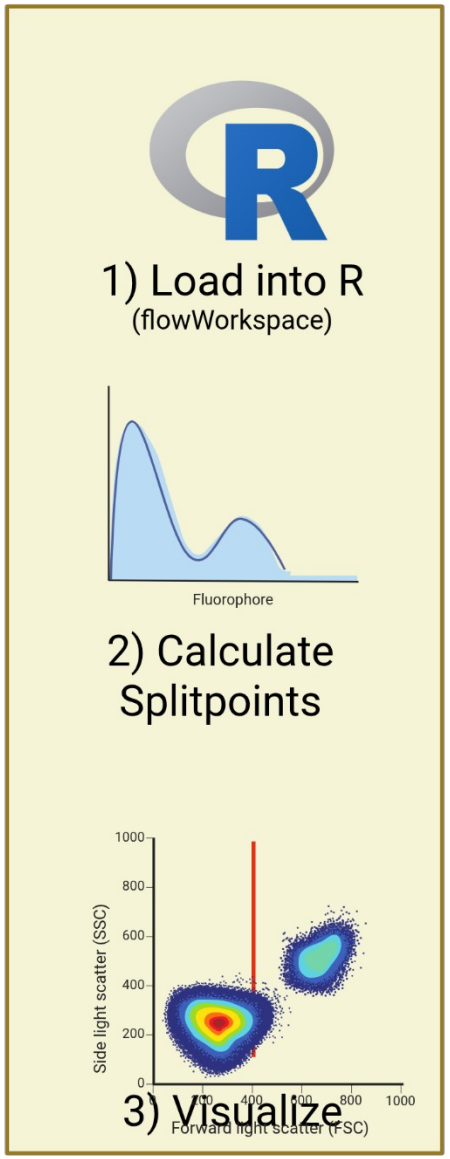
UNIVERSITY of MARYLAND SCHOOL OF MEDICINE

CC BY SA

Coereba



Code



UNIVERSITY of MARYLAND
SCHOOL OF MEDICINE

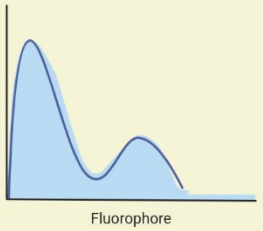
CC BY SA

Coereba

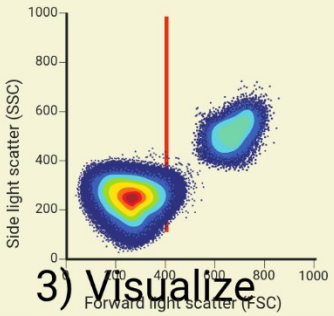


Code

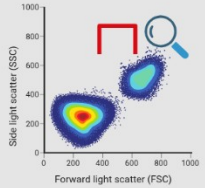
1) Load into R
(flowWorkspace)



2) Calculate Splitpoints

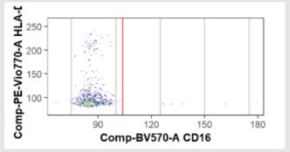


3) Visualize



4) Gate Constraints

5) Shiny App adjustments



6) Splitpoints updated

UNIVERSITY of MARYLAND
SCHOOL OF MEDICINE

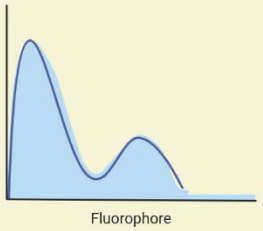
CC BY SA

Coereba

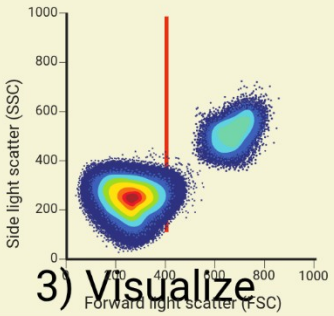


Code

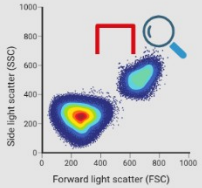
1) Load into R
(flowWorkspace)



2) Calculate Splitpoints

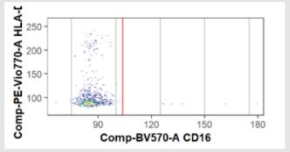


3) Visualize



4) Gate Constraints

5) Shiny App adjustments

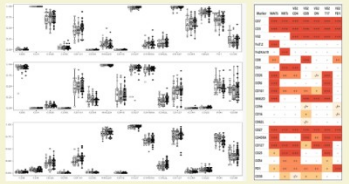


6) Splitpoints updated



7) CoerebaID

8) Summarized Experiment

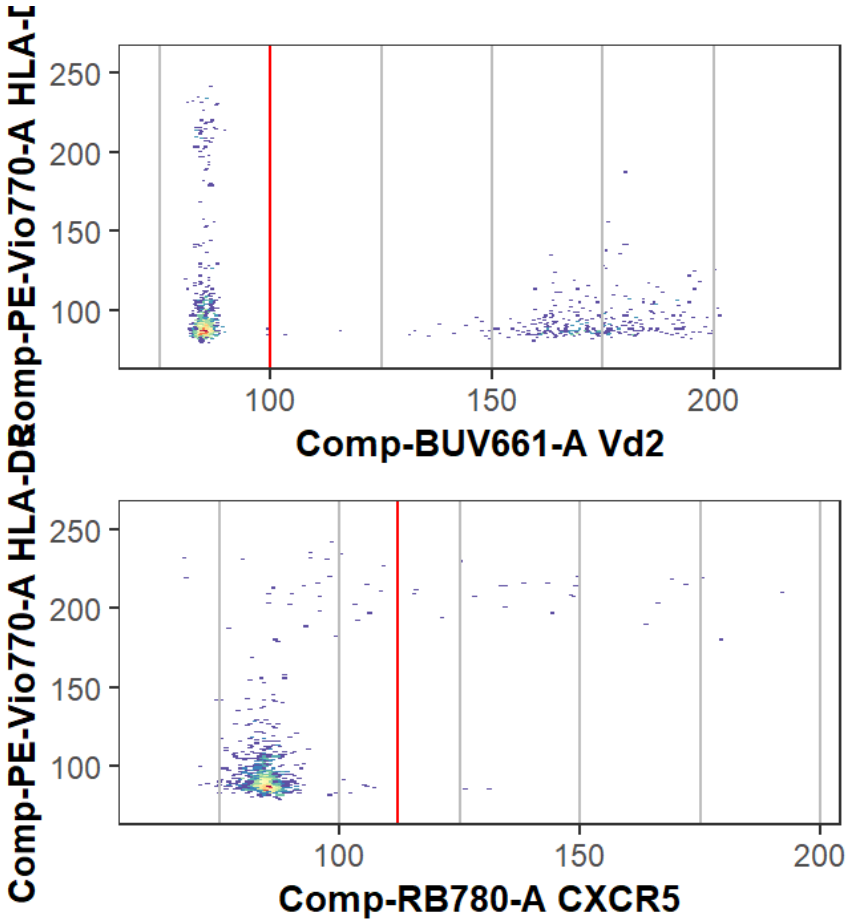


9) Visualize

UNIVERSITY of MARYLAND
SCHOOL OF MEDICINE

CC BY SA

Initial automated gate, with ability to adjust

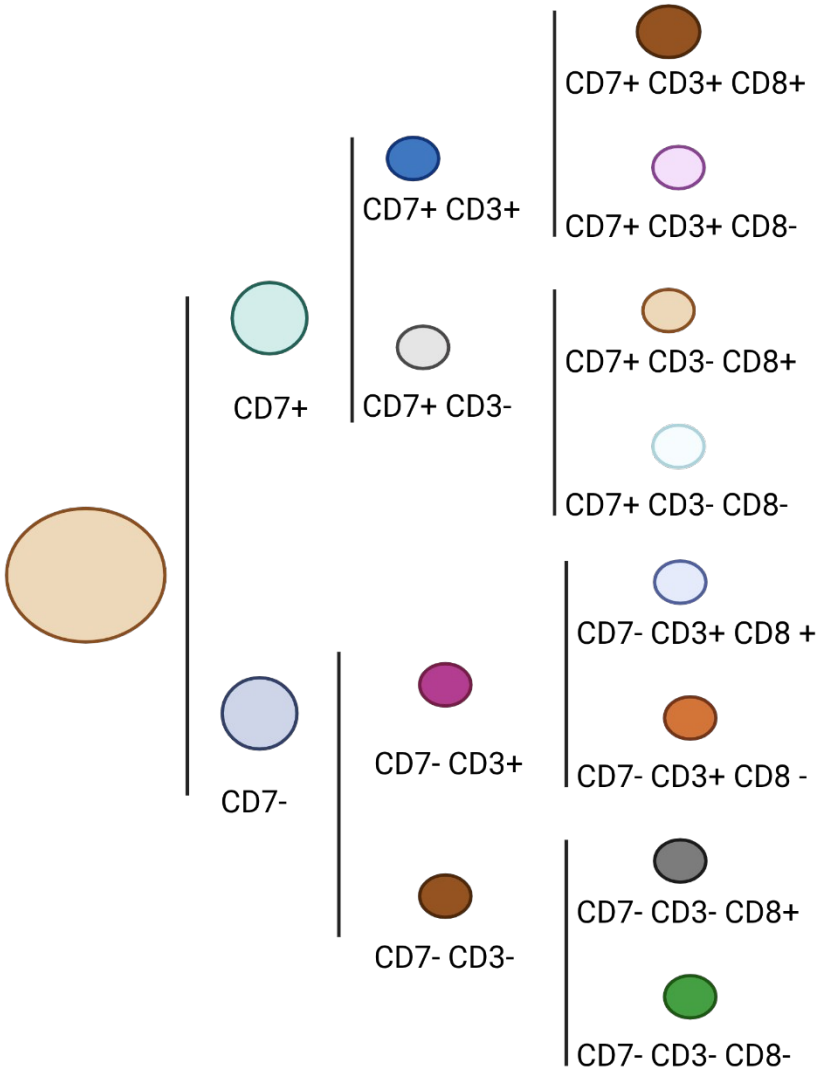


- ❑ Initial gates for designated markers calculated via openCyto (mindensity)
- ❑ Ability to update gate_range constraints
- ❑ Launch a Shiny app and designate interactively

CC BY SA

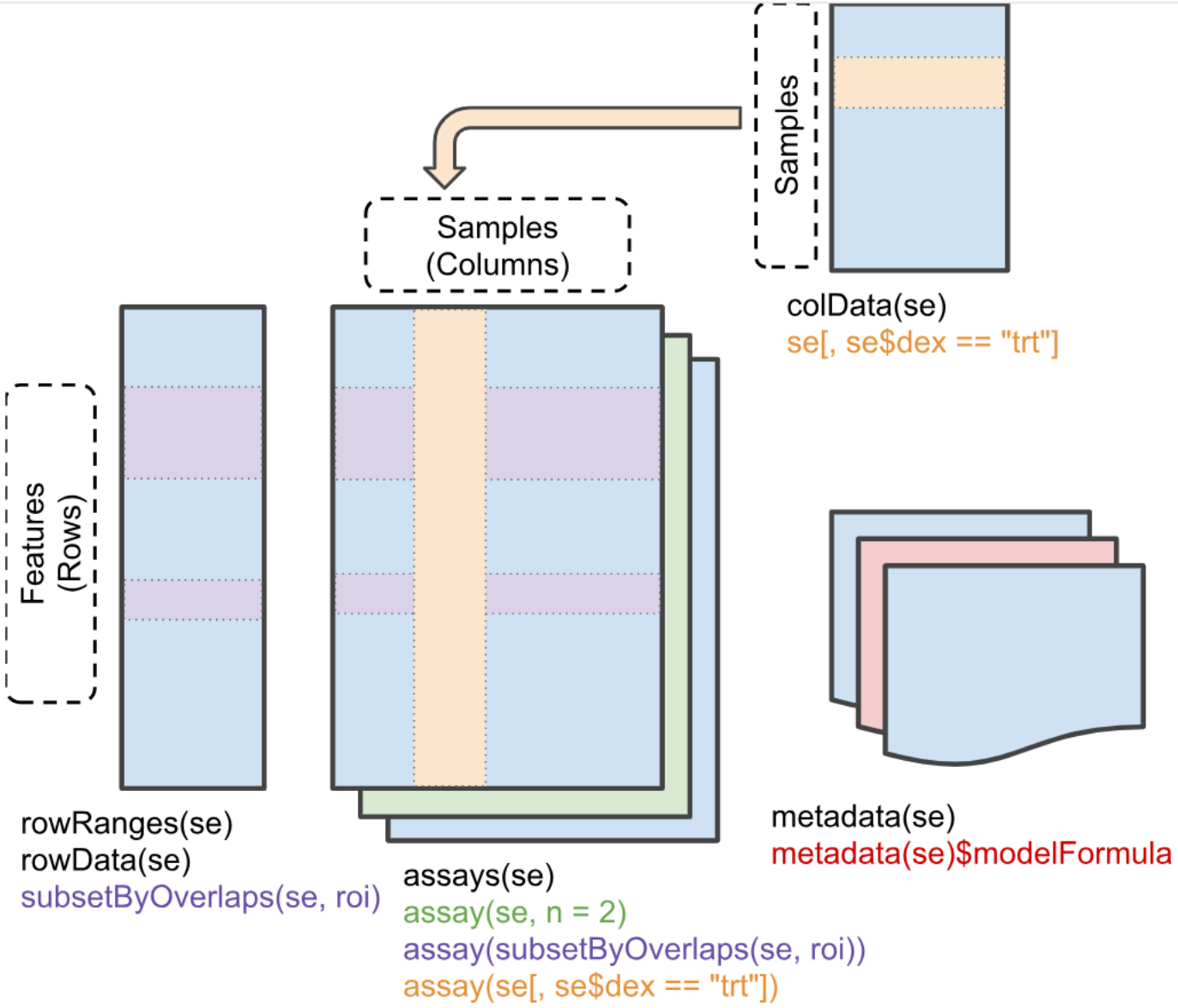
CoerebaID column gets added to the .fcs file

PEVio770pos	APCneg	AlexaFluor647neg	APCR700neg	APCFire750neg	APCFire810neg
PEVio770pos	APCneg	AlexaFluor647neg	APCR700neg	APCFire750neg	APCFire810neg
PEVio770pos	APCpos	AlexaFluor647neg	APCR700neg	APCFire750pos	APCFire810neg
PEVio770pos	APCneg	AlexaFluor647pos	APCR700neg	APCFire750neg	APCFire810pos



UNIVERSITY of MARYLAND
SCHOOL OF MEDICINE

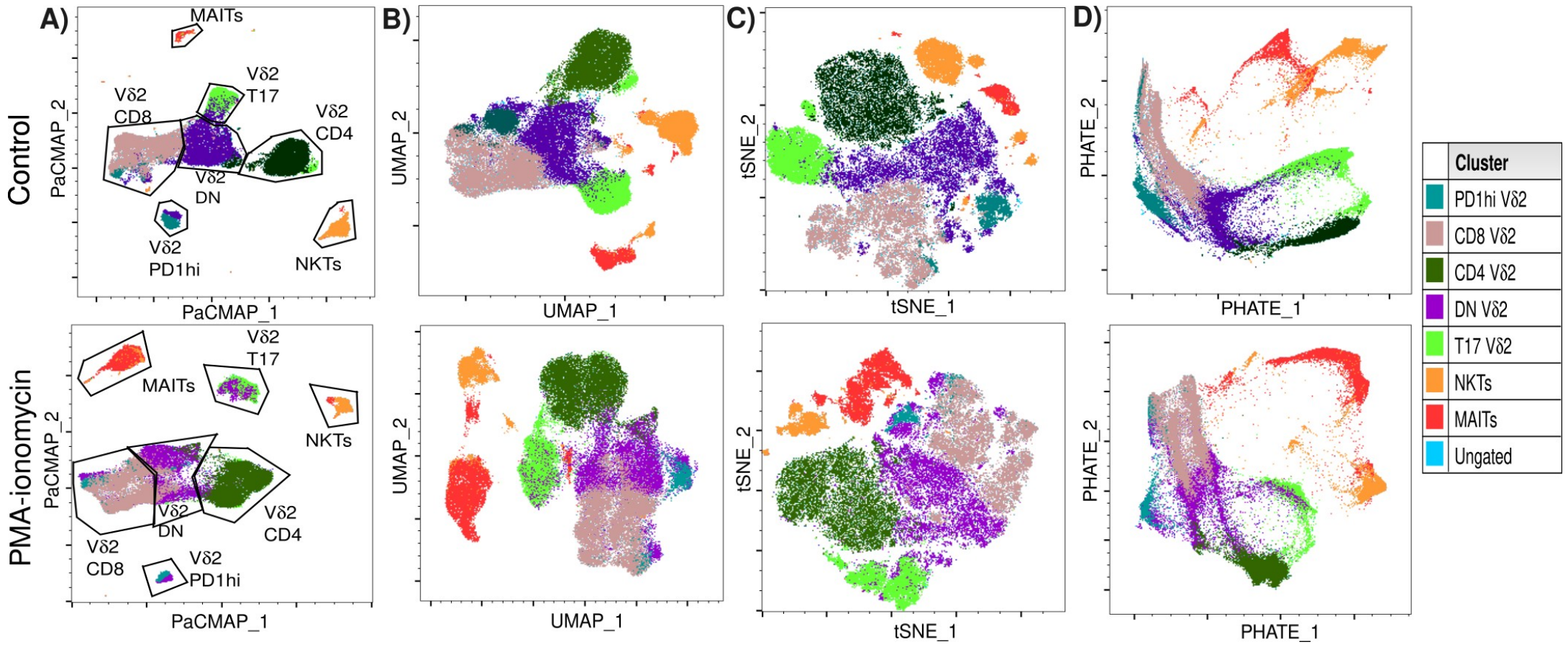
Summarized Experiment



- Permits rapid re-aggregation of the terminal nodes for re-evaluation
- Ties in with downstream Bioconductor statistics packages
- Portable. Don't like how we aggregated groups of cells? File is small enough to store in the same data repository as .fcs files for subsequent re-analysis.
- Filterable by metadata or node.
  - Nodes only in HEU-hi?
  - Nodes unique to one individual?

CC BY SA

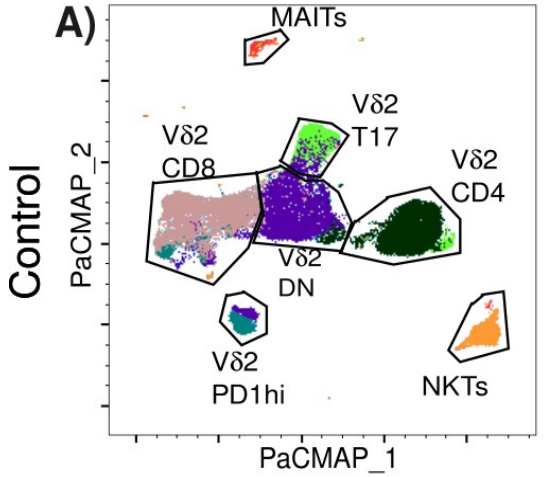
Ability to contrast unsupervised methods vs. “manual” ground-truth



UNIVERSITY of MARYLAND
SCHOOL OF MEDICINE

CC BY SA

Retrieving and re-enumerating

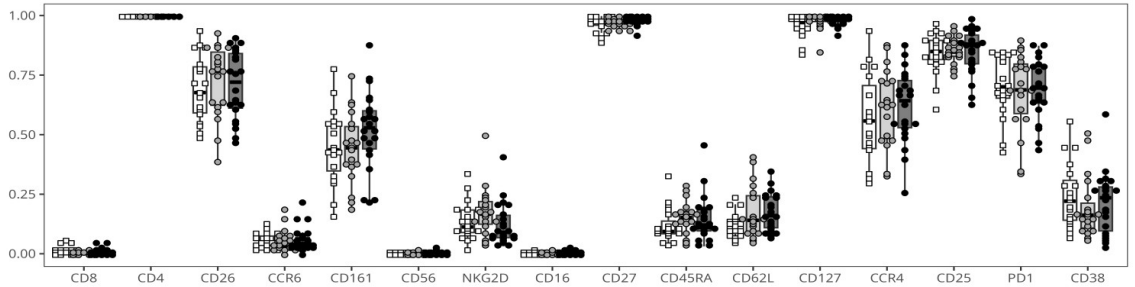


1) Gate

2) Retrieve manual-annotated gates

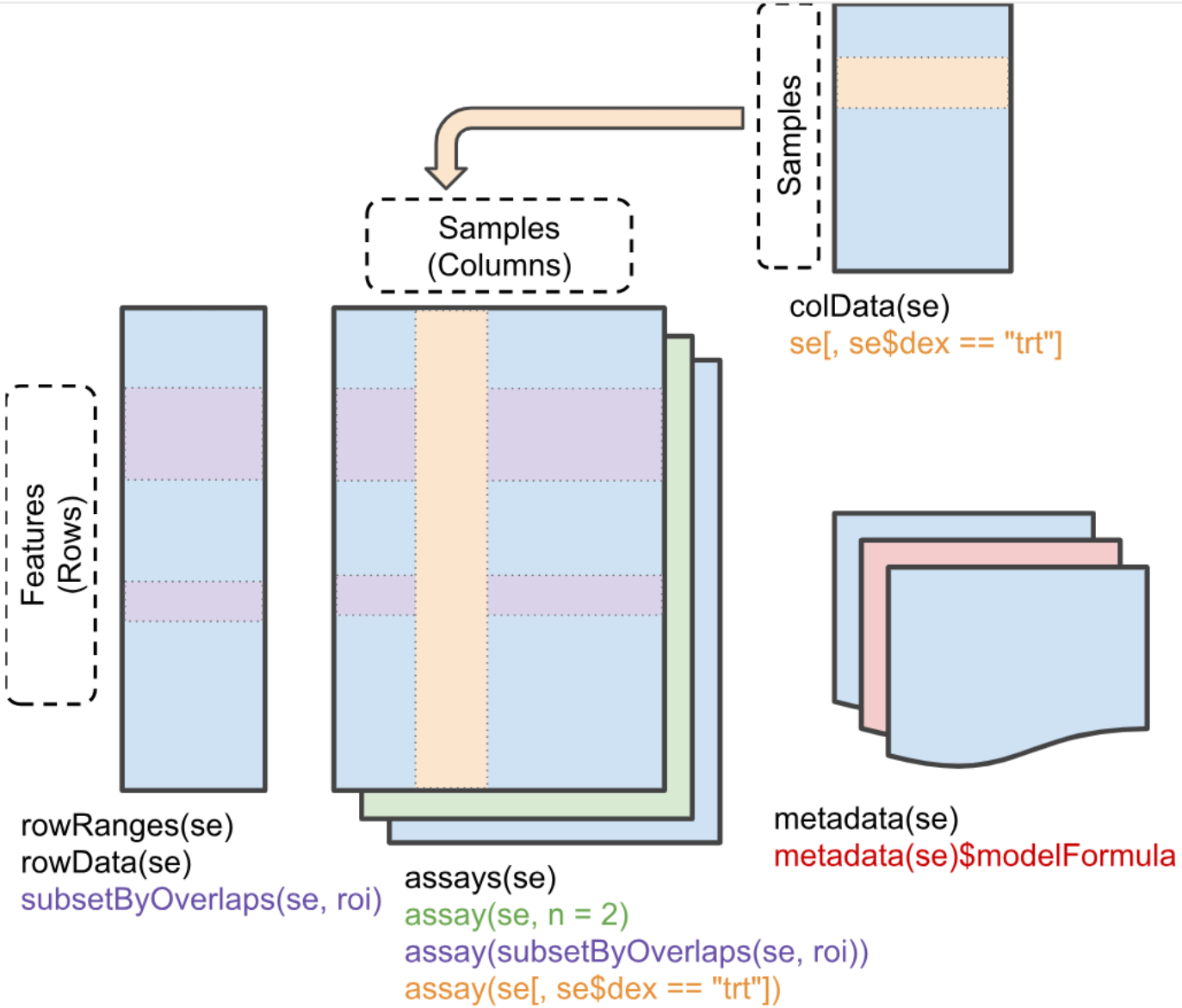
PEVio770pos	APCneg	AlexaFluor647neg	APCR700neg	APCFire750neg	APCFire810neg
PEVio770pos	APCneg	AlexaFluor647neg	APCR700neg	APCFire750neg	APCFire810neg
PEVio770pos	APCpos	AlexaFluor647neg	APCR700neg	APCFire750pos	APCFire810neg
PEVio770pos	APCneg	AlexaFluor647pos	APCR700neg	APCFire750neg	APCFire810pos

3) Re-enumerate and plot



CC BY SA

Summarized Experiment



- Permits rapid re-aggregation of the terminal nodes for re-evaluation
- Ties in with downstream Bioconductor statistics packages
- Portable. Don't like how we aggregated groups of cells? File is small enough to store in the same data repository as .fcs files for subsequent re-analysis.
- Filterable by metadata or node.
  - Nodes only in HEU-hi?
  - Nodes unique to one individual?

CC BY SA

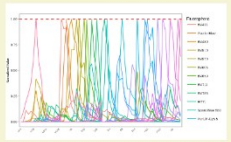
Semi-supervised Pipeline



1) Acquisition
(Cytek Aurora 5L)



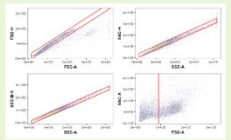
2) Unmixing Controls
(Luciernaga)



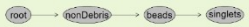
3) Unmixing
(OLS, TRU-OLS, AutoSpectral)



4) File Checks
(PeacoQC, Flow AI, etc)



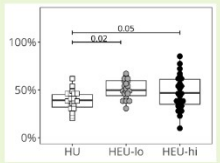
5) Clean Up Gates
(FlowJo or flowWorkspace)



6) Initial Gates
(openCyto or flowMagic)



7) FlowJo .wsp
(CytoML Docker)



8) Major subset Outputs

Iterating by Experiment



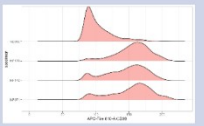
9) Subset Export



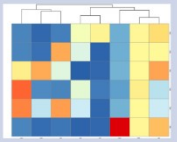
10) Return to R
(flowWorkspace)



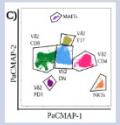
11) Coereba



12) Normalization
(CytoNorm or CyCombine)



13) Clustering
(FlowSOM)



14) Dimensionality Visualization
(UMAP or PaCMAP)



Summarized Experiment



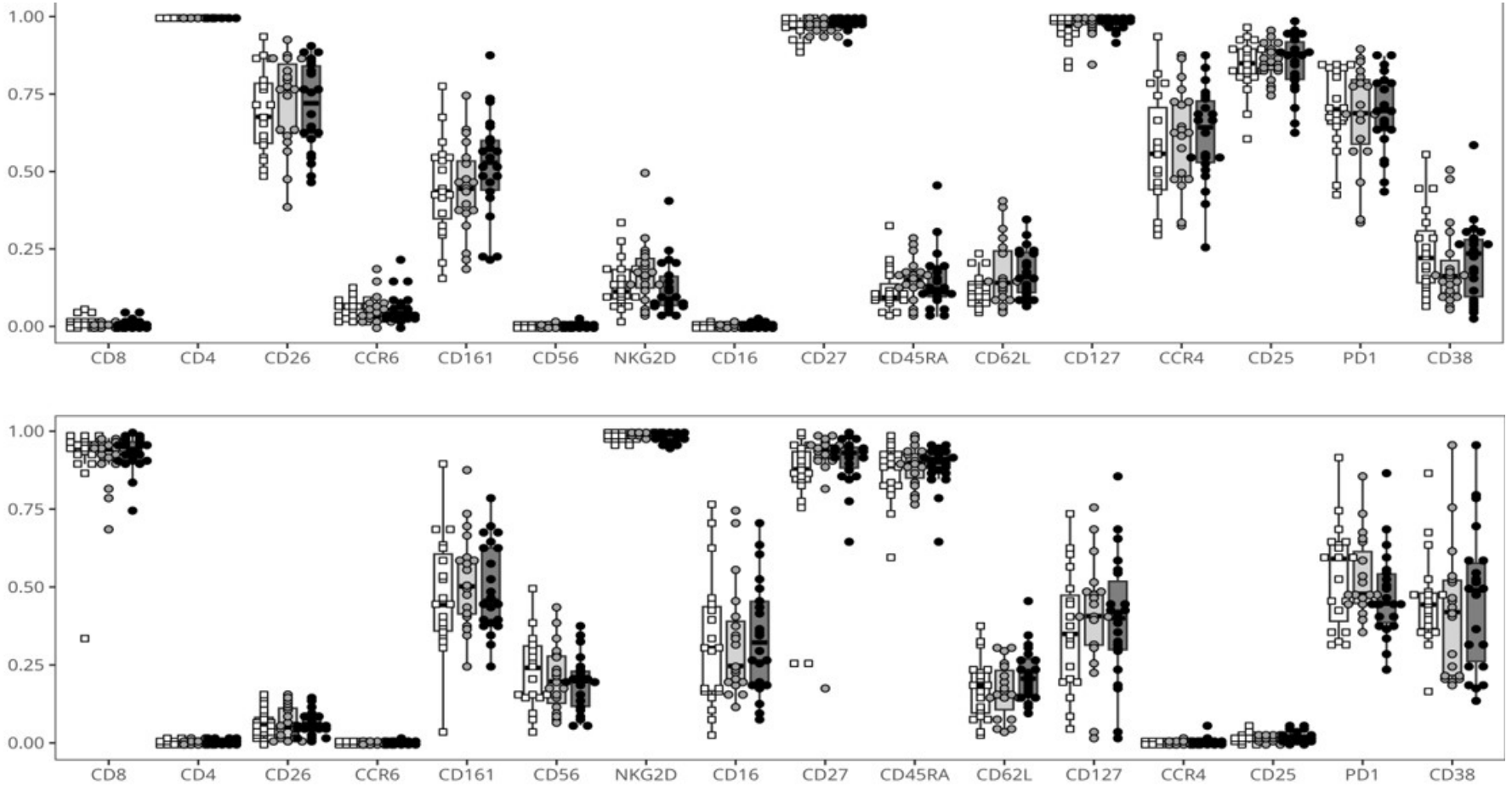
15) Coereba

Iterating by Cell Subset

CC BY SA



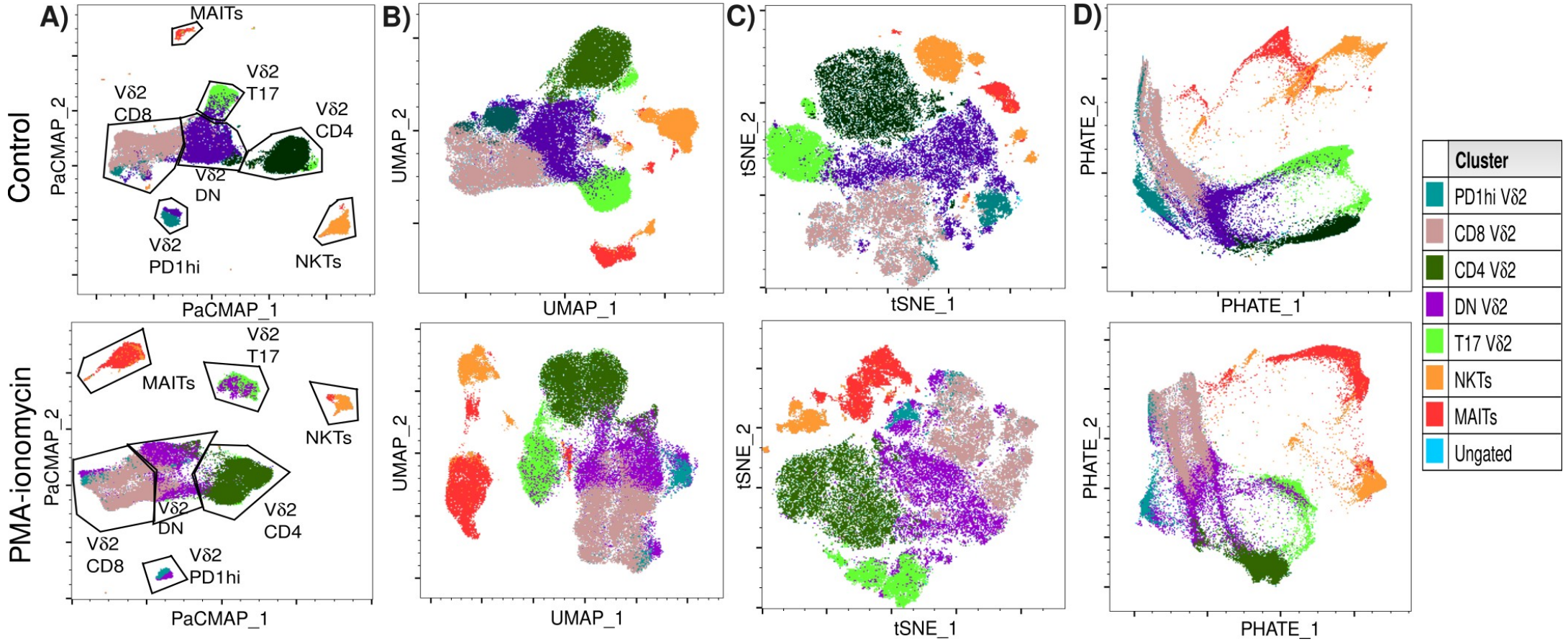
Retrieval of marker expressions



UNIVERSITY of MARYLAND
SCHOOL OF MEDICINE

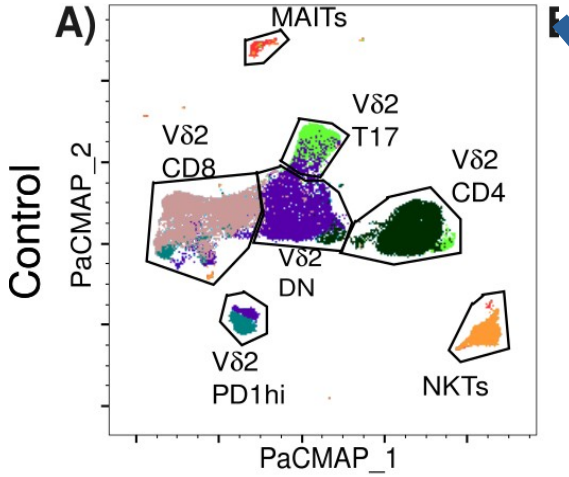
CC BY SA

Ability to contrast unsupervised methods vs “manual” ground-truth



CC BY SA

Retrieving and re-enumerating

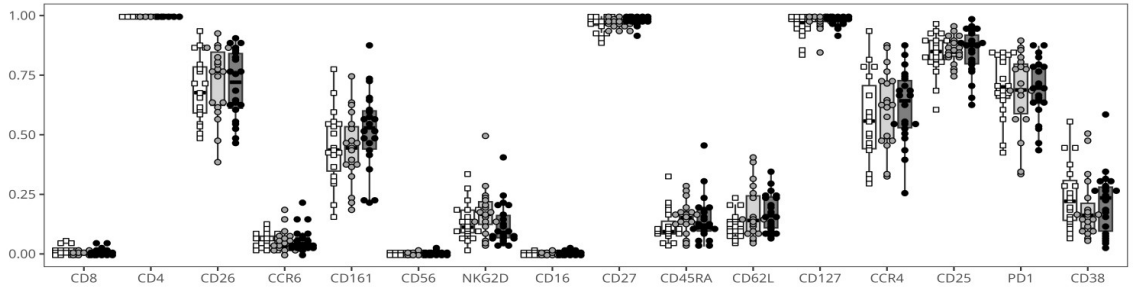


1) Gate

2) Retrieve manual-annotated gates

PEVio770pos	APCneg	AlexaFluor647neg	APCR700neg	APCFire750neg	APCFire810neg
PEVio770pos	APCneg	AlexaFluor647neg	APCR700neg	APCFire750neg	APCFire810neg
PEVio770pos	APCpos	AlexaFluor647neg	APCR700neg	APCFire750pos	APCFire810neg
PEVio770pos	APCneg	AlexaFluor647pos	APCR700neg	APCFire750neg	APCFire810pos

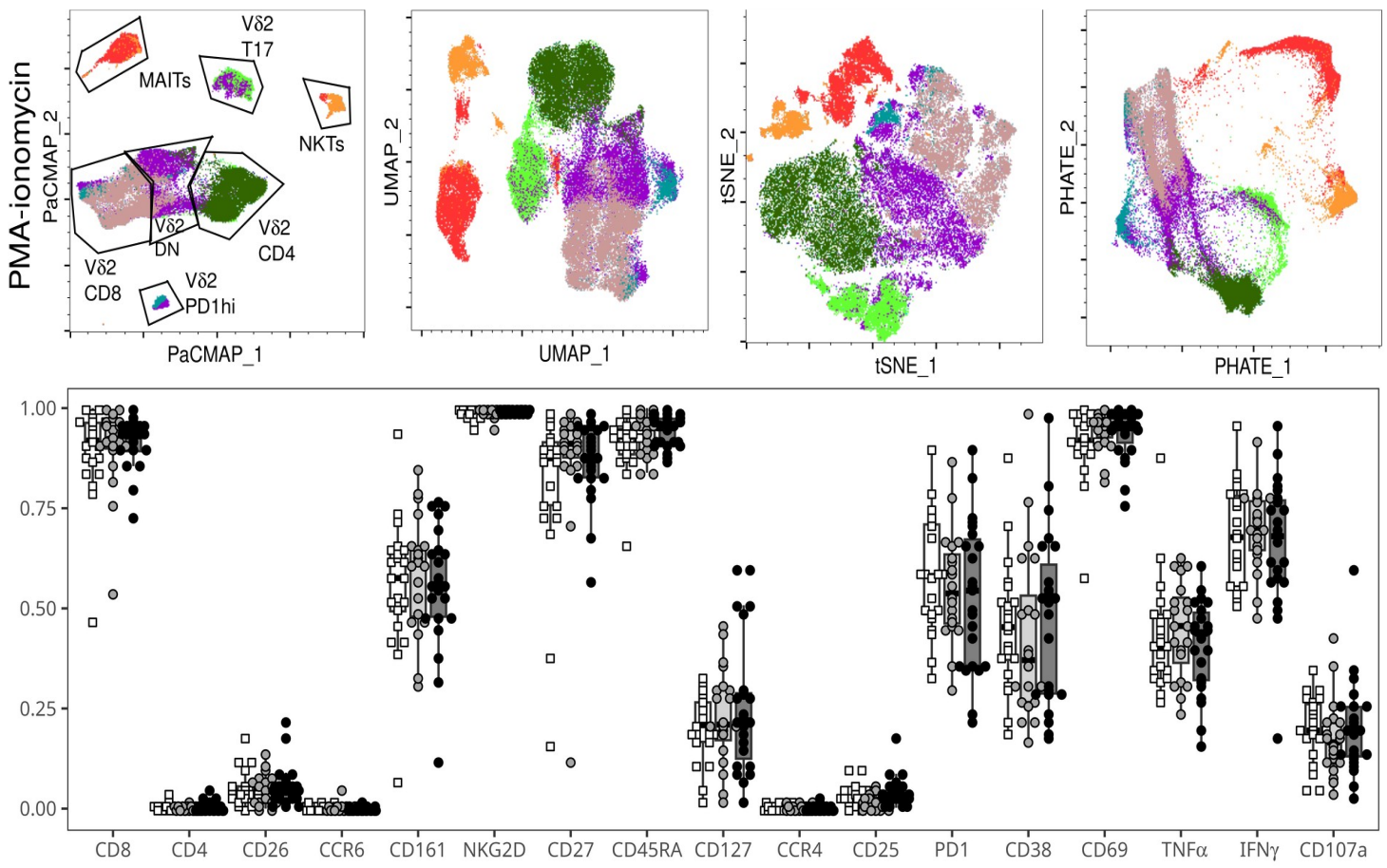
3) Re-enumerate and plot



CC BY SA

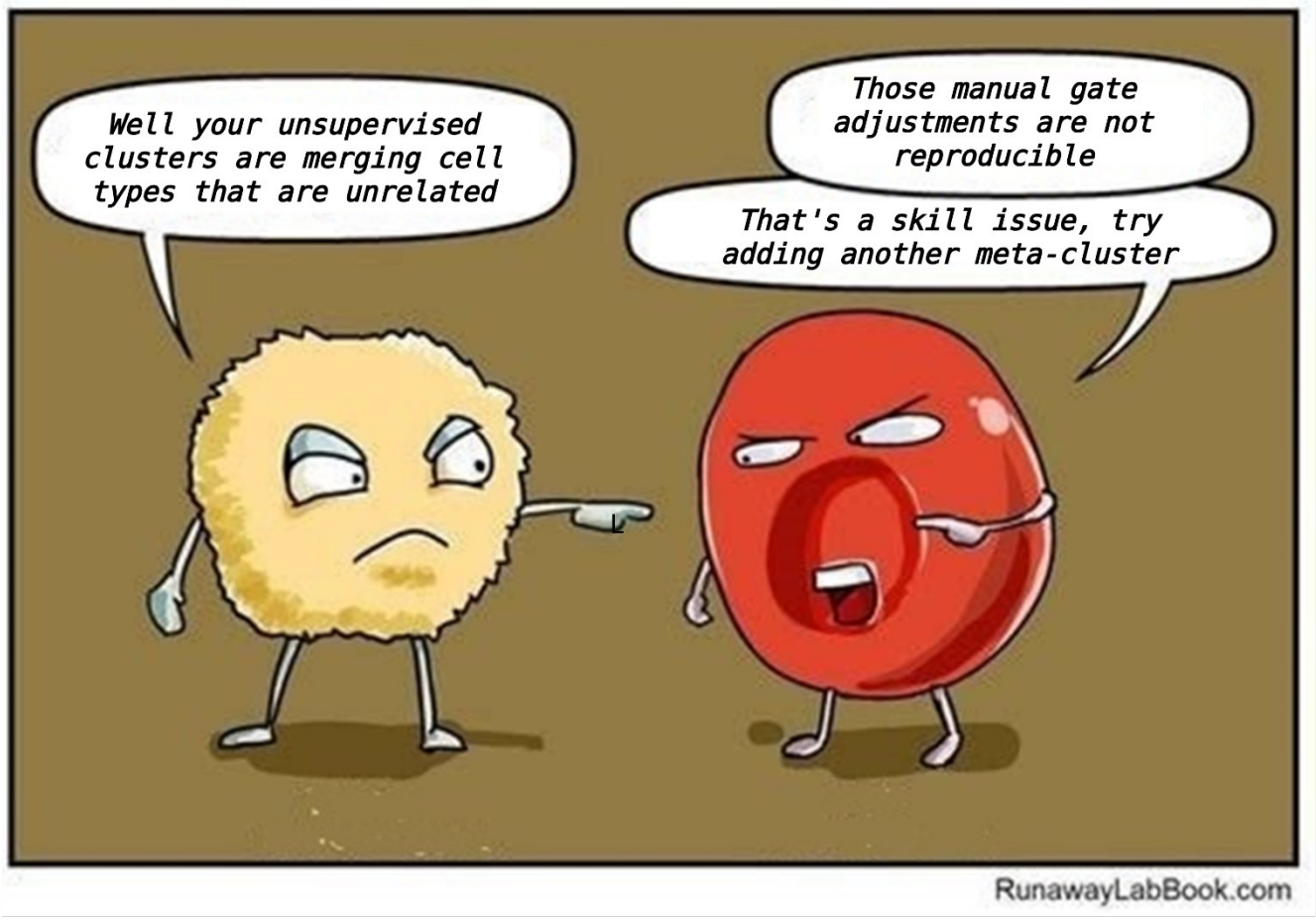
Retrieve manual annotated gating information following unsupervised steps

Marker	MAITs	NKTs	Vδ2 CD4	Vδ2 CD8	Vδ2 DN	Vδ2 T17	Vδ2 PD1
CD7	+++	+++	+++	+++	+++	+++	+++
CD3	+++	+++	+++	+++	+++	+++	+++
Vδ2	-	-	+++	+++	+++	+++	+++
Vα7.2	+++	-	-	-	-	-	-
Vα24Jα18	-	+++	-	-	-	-	-
CD8	++	-	-	+++	-	-	++
CD4	-	+++	+++	-	-	-	-
CD26	+++	++	++	-	-/+	+++	-
CCR6	+++	-	-	-	-	+++	-
CD161	+++	++	++	++	+	+++	-
NKG2D	+++	-	-	+++	+++	+++	-
CD56	-	-	-	-/+	-	-	-/+
CD16	-	-	-	+	-	-	-/+
CD62L	-	-	-	-/+	-	-	-
CD27	+++	+++	+++	+++	+++	+++	+++
CD45RA	+++	-	-	+++	++	+++	+++
CD127	+++	+++	+++	+	+++	+++	-
CD25	+	+++	+++	-	-	+++	-
CCR4	-	++	++	-	-	+	-
PD1	+	++	++	++	++	+	+++
CD38	-	+	-/+	+	+	-	-



CC BY SA

Functional and Iterative: plenty for everyone to dislike :)

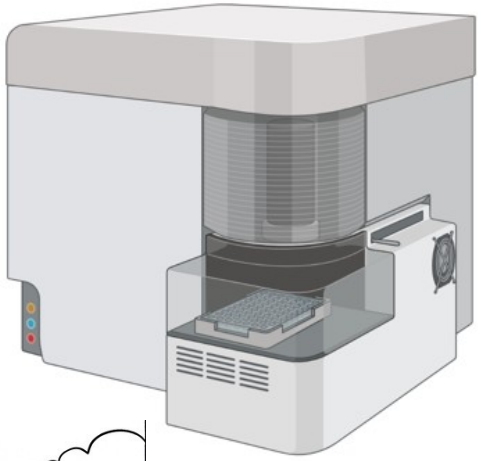


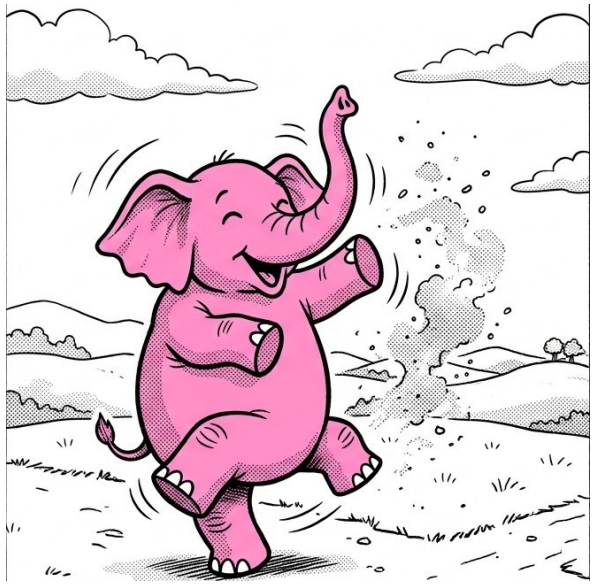
Fun off-label uses thus far:

- Contrasting where “manually-gated” cells end up vs. our unsupervised methods (normalization, clustering, etc.)
- Added benefit of looking-at-the-forest: Easy to identify badly cryopreserved specimens where cell populations were lost

CC BY SA

To unlock the comprehensive biological discovery potential, we need to address these pink elephants



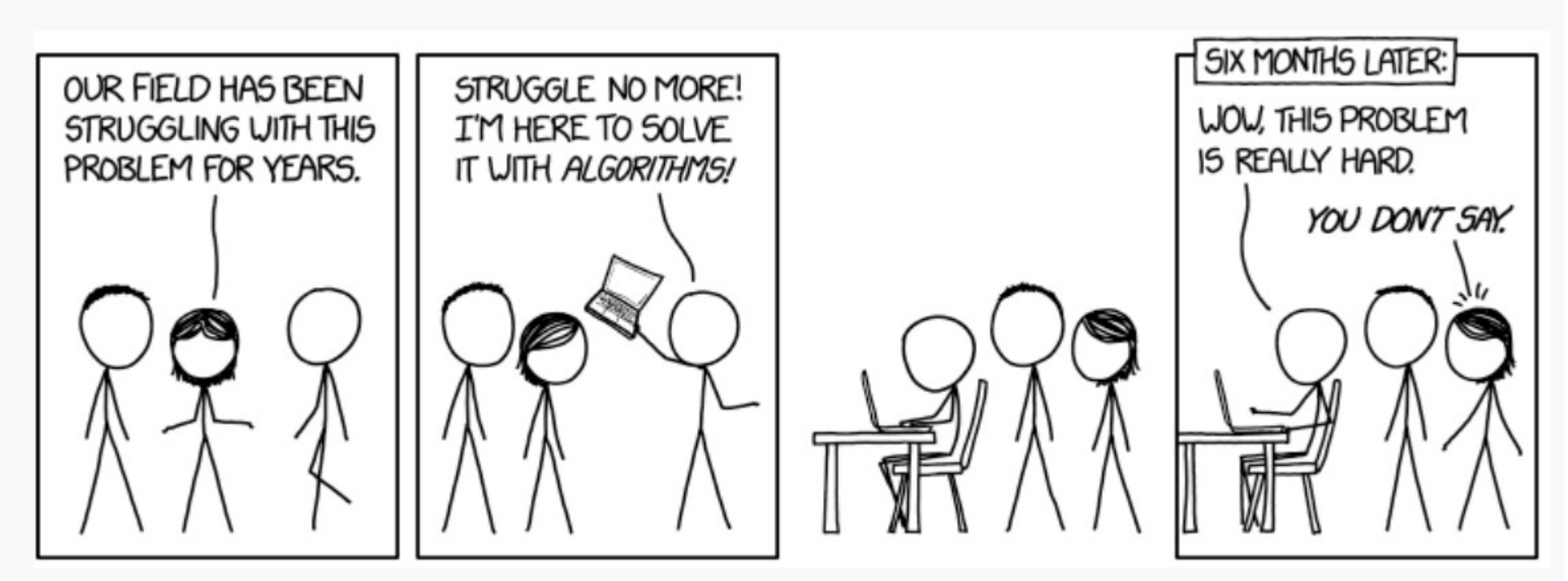


- Quality of the unmixed full-stained data is linked to the quality of our single-color controls (and our ability to provide appropriate autofluorescence controls).

- If we are acquiring 3 million cells to detect 100 rare cells, how do we thoroughly explore the rest of the sample to avoid wasting useful data?

CC BY SA

Are our high-dimensional analysis problems fixed?



CC BY SA

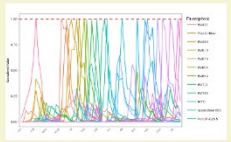
Semi-supervised Pipeline



1) Acquisition
(Cytek Aurora 5L)



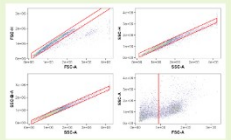
2) Unmixing Controls
(Luciernaga)



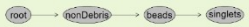
3) Unmixing
(OLS, TRU-OLS, AutoSpectral)



4) File Checks
(PeacoQC, Flow AI, etc)



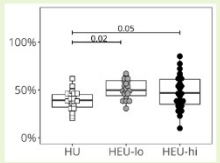
5) Clean Up Gates
(FlowJo or flowWorkspace)



6) Initial Gates
(openCyto or flowMagic)



7) FlowJo .wsp
(CytoML Docker)



8) Major subset Outputs

Iterating by Experiment



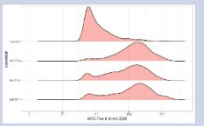
9) Subset Export



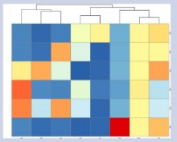
10) Return to R
(flowWorkspace)



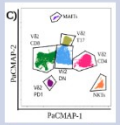
11) Coereba



12) Normalization
(CytoNorm or CyCombine)



13) Clustering
(FlowSOM)



14) Dimensionality Visualization
(UMAP or PaCMAP)



Summarized Experiment



15) Coereba

Iterating by Cell Subset

CC BY SA

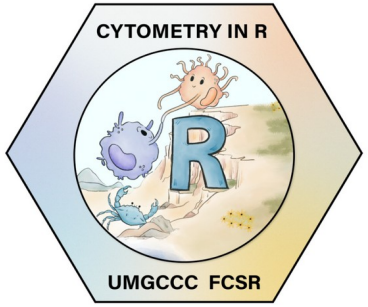
What good is any tool if the community is not able to use it?

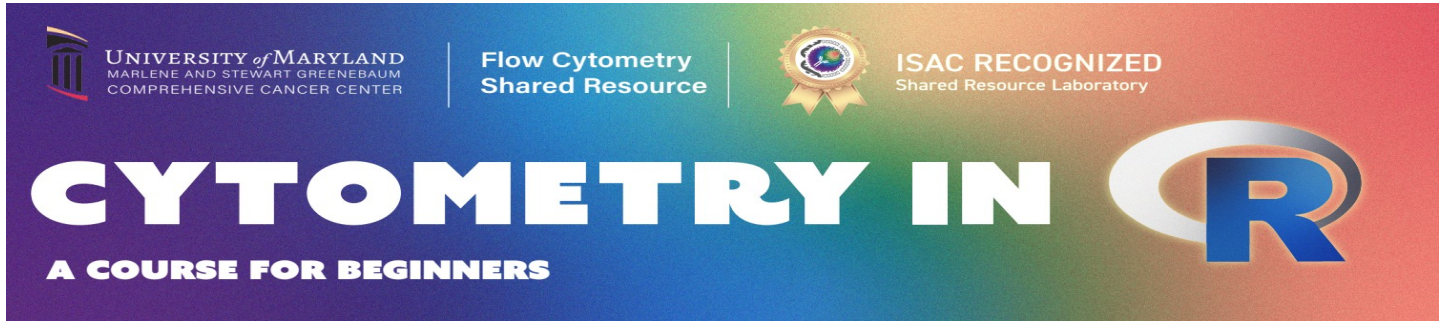
- Few resources to learn Cytometry in R to begin with
- Those that exist aimed at those with intermediate-advanced R skills, not beginners
- Know-how concentrated at a few institutions
- Most self-learning attempts end in frustration



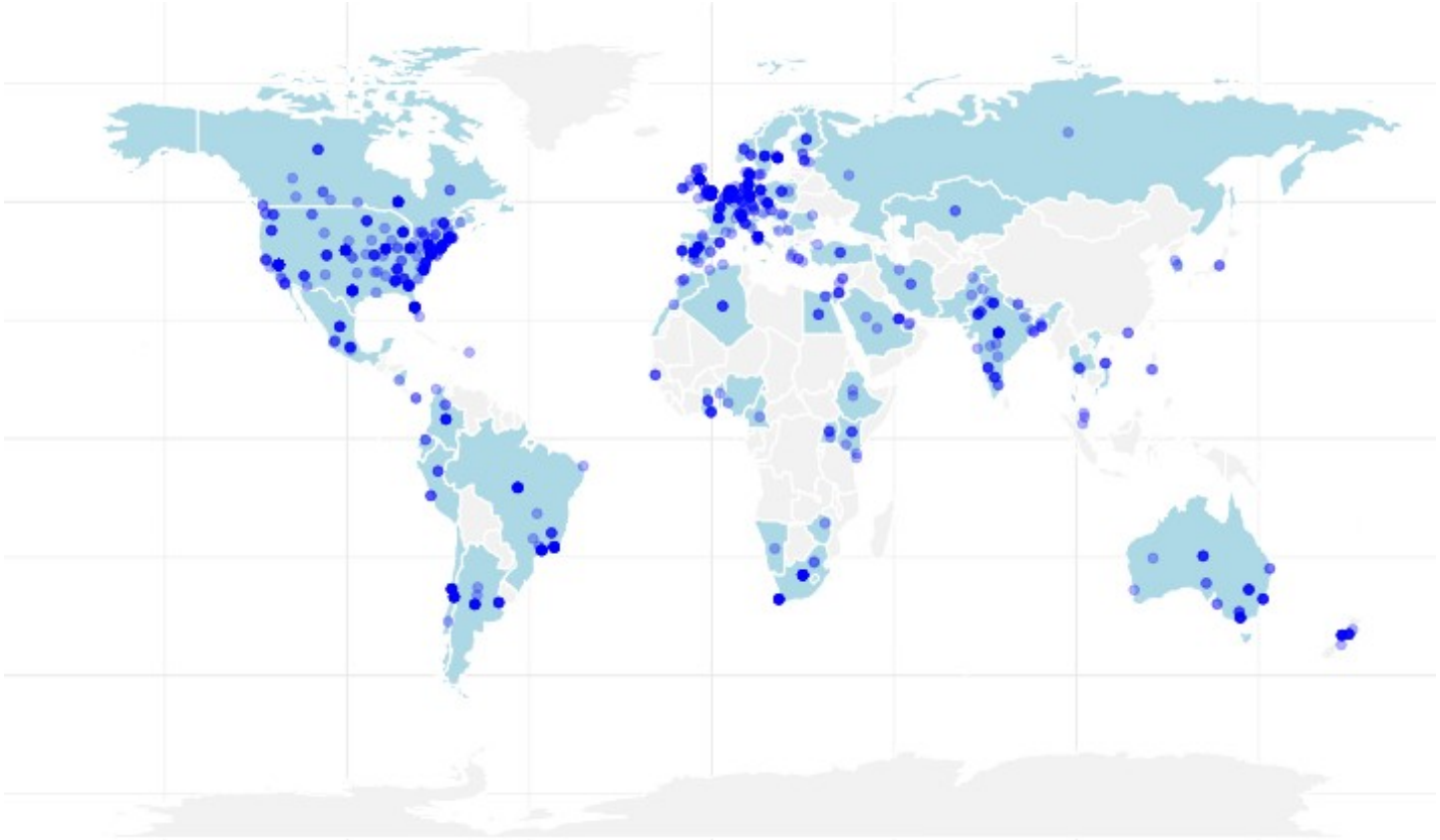
CC BY SA

Cytometry in R: A free weekly course for flow cytometrist with no-to-little coding experience





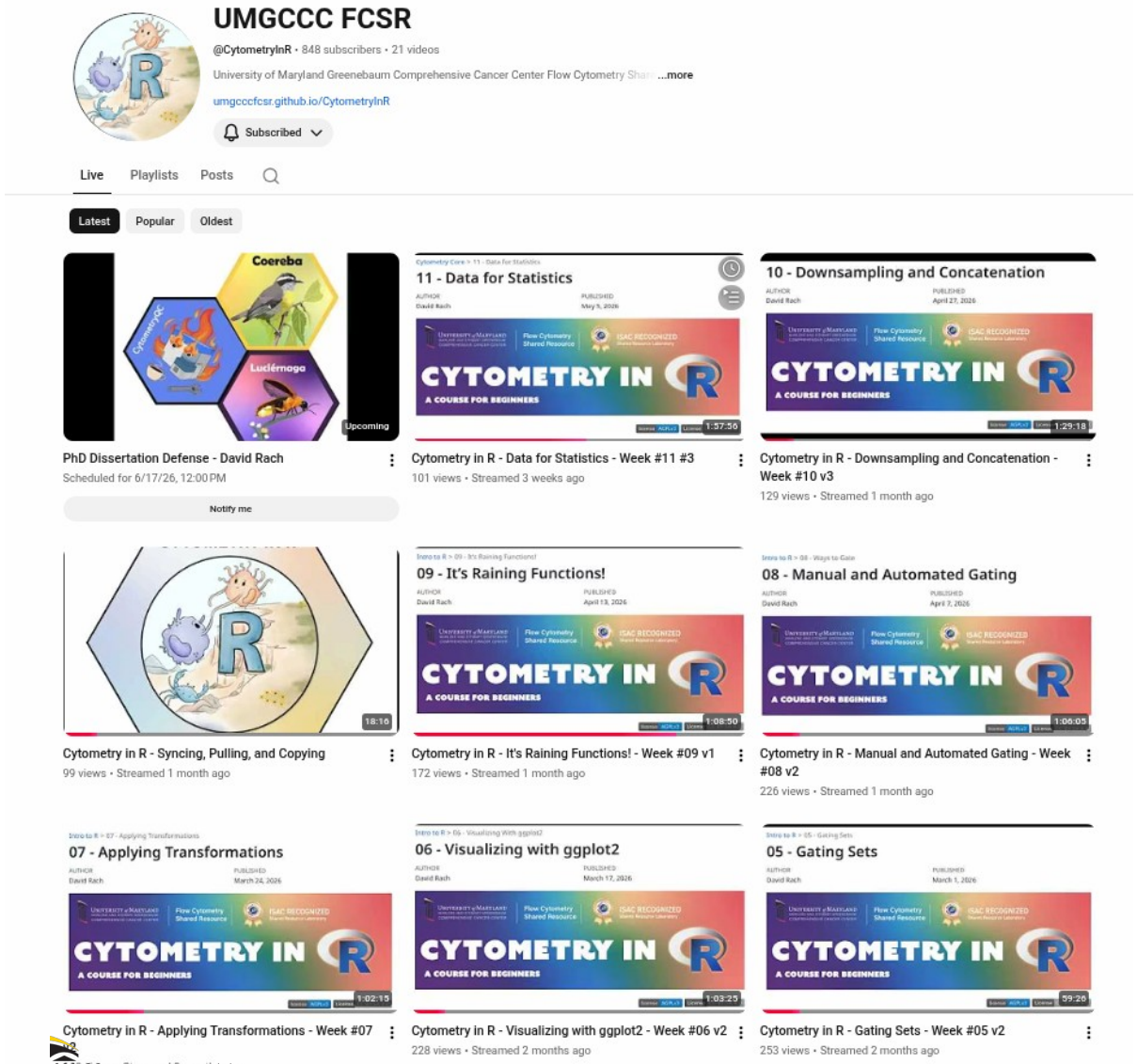
Website



UNIVERSITY of MARYLAND
SCHOOL OF MEDICINE

CC BY SA

YouTube

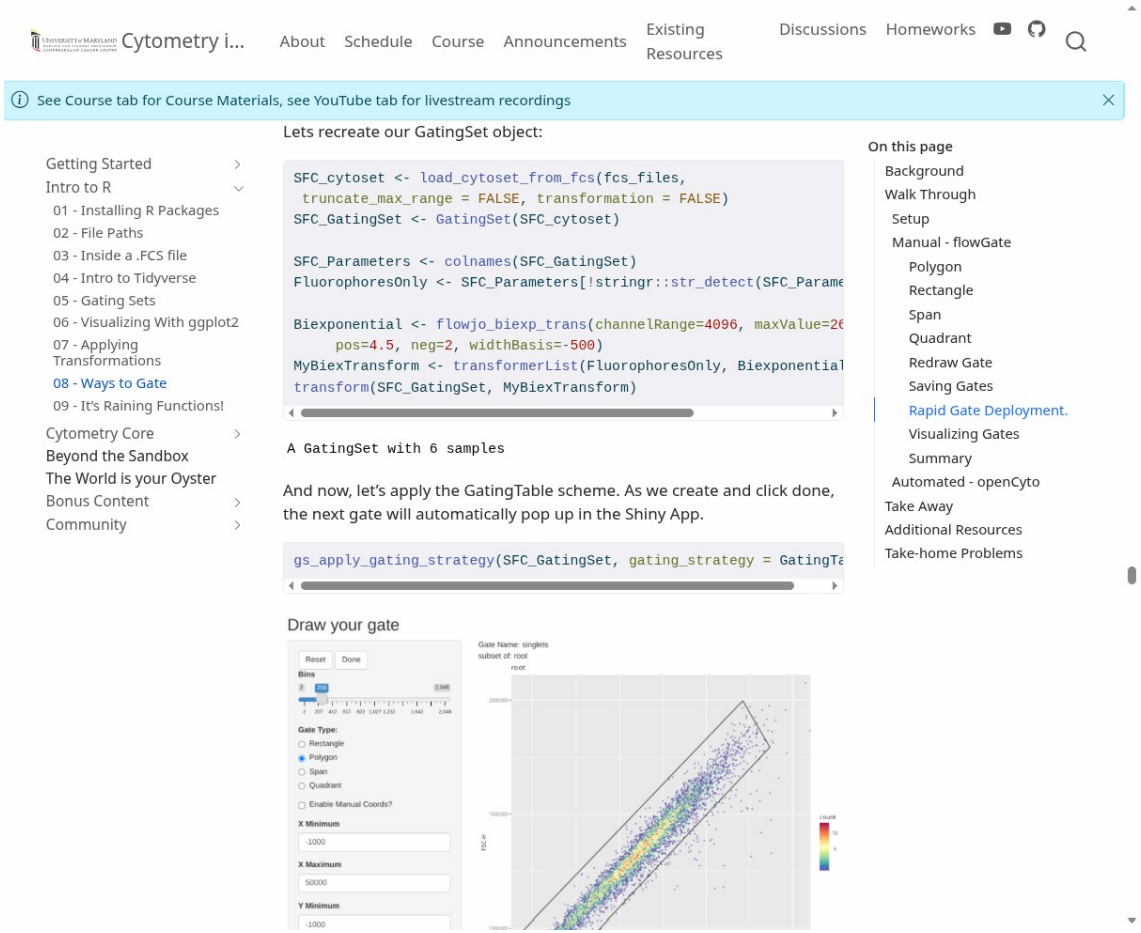


- Three live-streams a week (accommodating different timezones and schedules)
- Recordings available immediately after
- Best of the weekly livestreams added to the playlist, cutting down on editing

CC BY SA

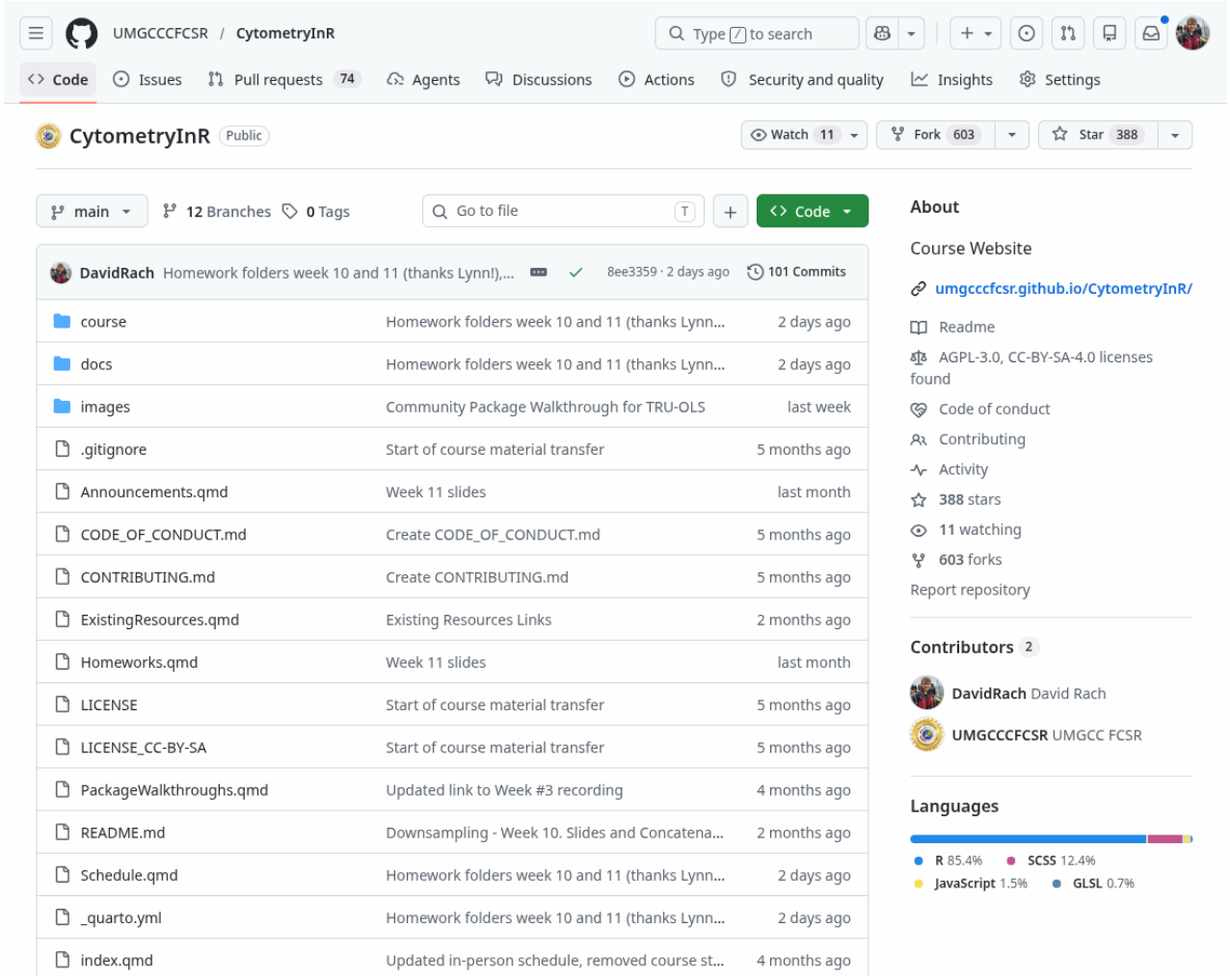
Website

- Created with Quarto, hosted via GitHub pages.
- Code, code outputs, screenshots and explanations
- Searchable
- New weekly material added as a new .qmd file



CC BY SA

GitHub

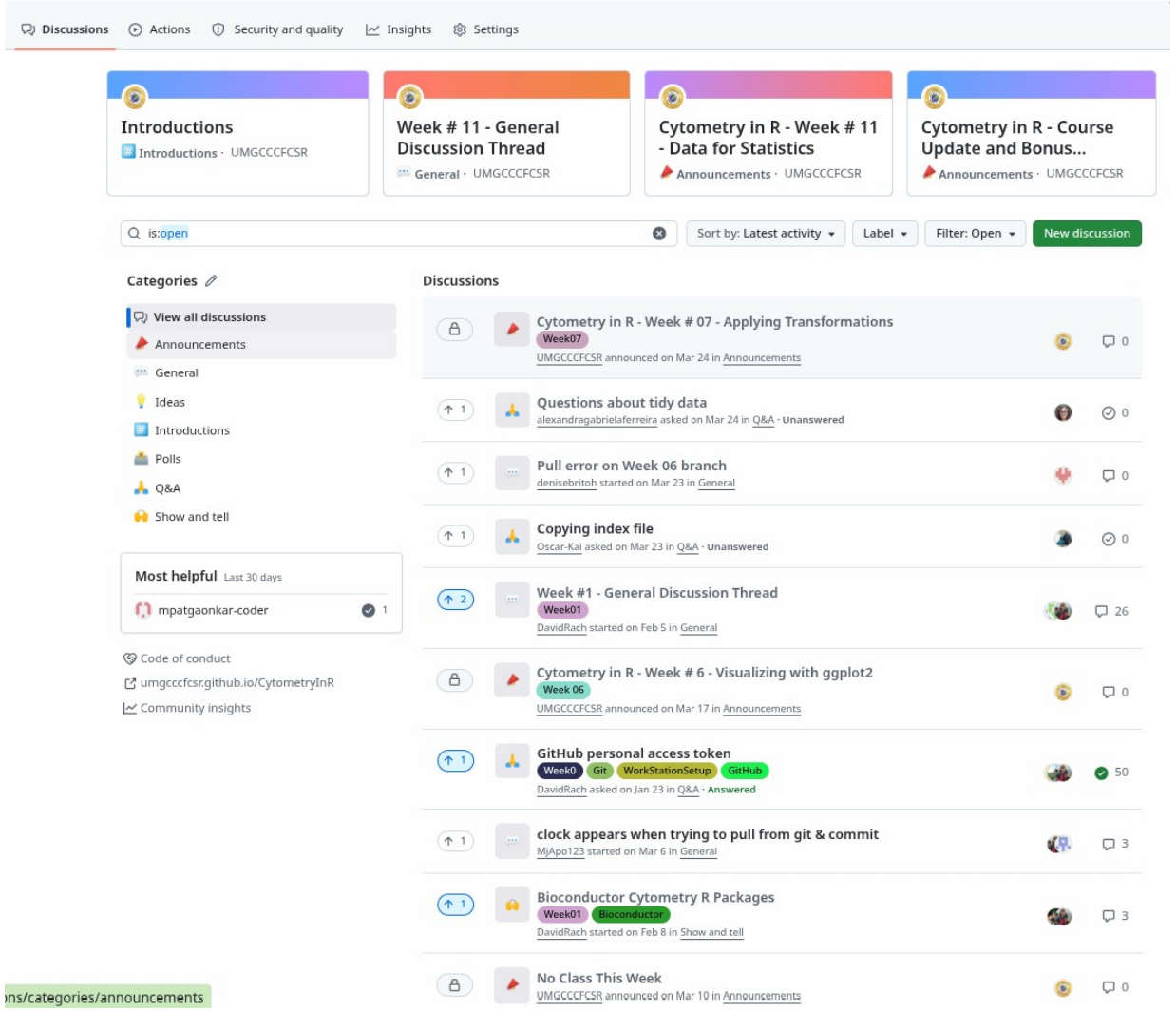


- Code and datasets available via the GitHub repository
- Course participants start course by creating GitHub account and forking the repository
- After Positron is linked to GitHub account, pull locally
- When new material posted each week, sync and pull

CC BY SA

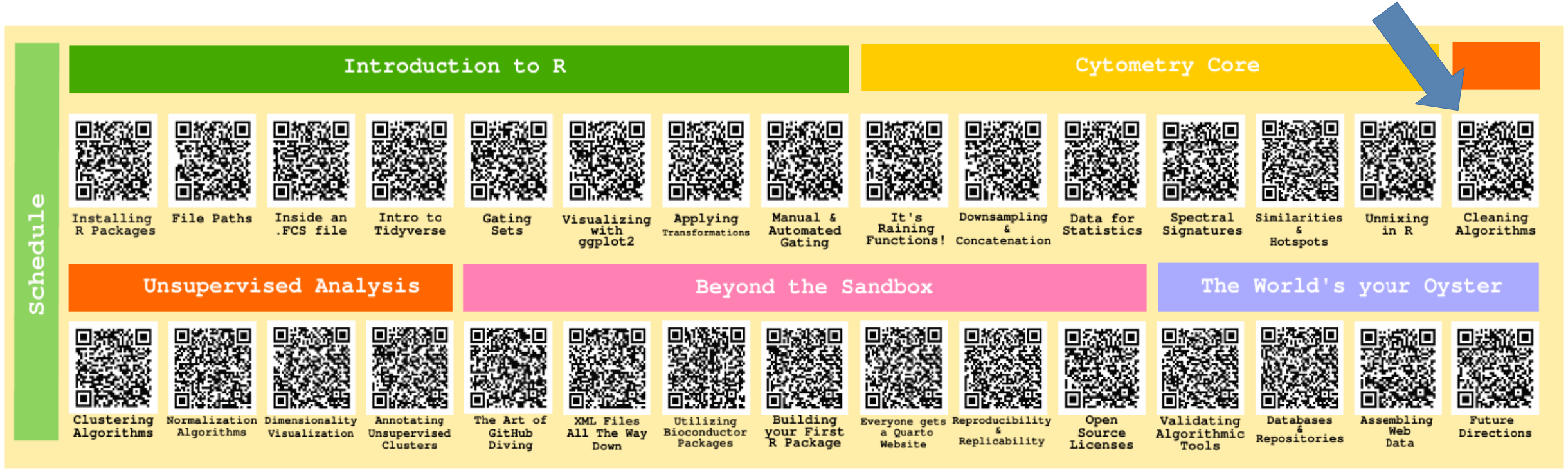
Discussion Forum

- GitHub Discussions page has been repurposed as a community forum for questions, etc.



CC BY SA

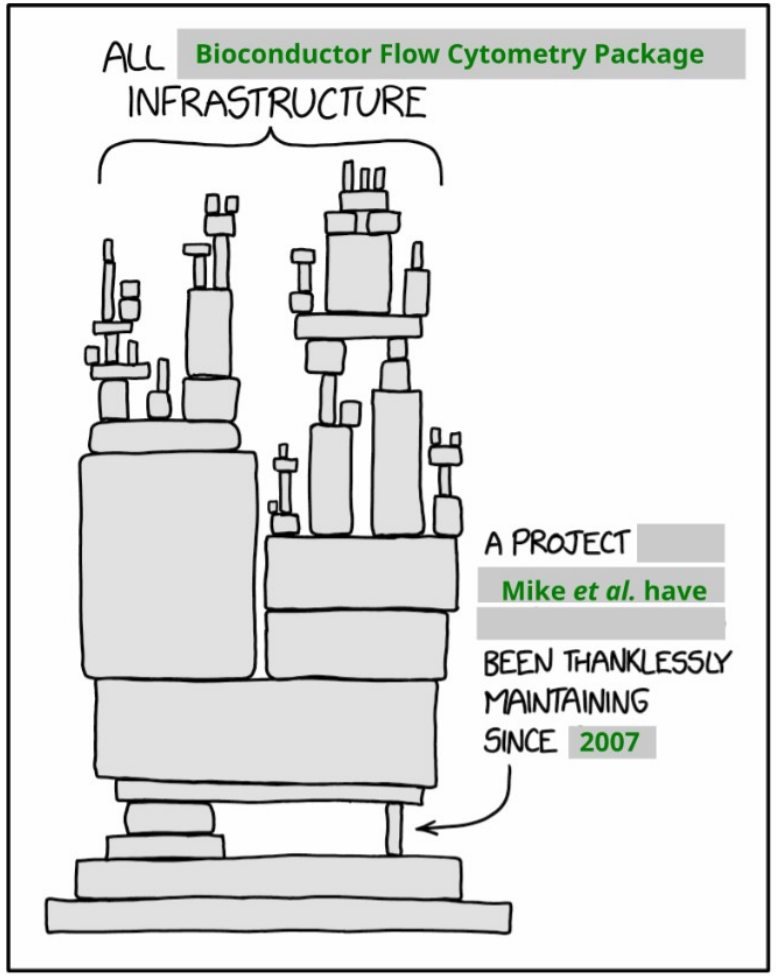
Half-way through this passion project



- Wrapping up Cytometry Core module
- Gathering steam to start on the Unsupervised Analysis section

CC BY SA

What good is any tool if the community is not able to use it?



- Working towards formally submitting my R packages to Bioconductor this year (adding documentation, generalizing for different instrument manufacturers)
- Still learning better ways to parallelize and speed up the process

- 72 Bioconductor cytometry packages (+ 20? GitHub based packages)
- Heavily under-utilized, some unmaintained
- Evidence that “Build the `software tool` and they will come” approach doesn’t work in practice

CC BY SA

UNIVERSITY of MARYLAND
SCHOOL OF MEDICINE

Acknowledgments

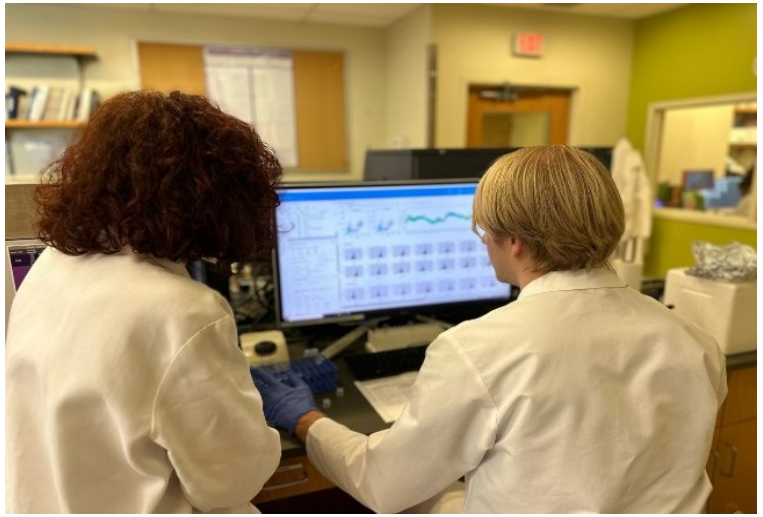
KAMUZU UNIVERSITY
OF HEALTH SCIENCES

All mothers and their infants, and their communities

All the open-source contributors, developers, and maintainers, that collectively made the tool sets without which all of this would not be feasible

Dr. Cristiana Cairo
Dr. Kirsten E. Lyke
Dr. Xiaoxuan Fan

Dr. Daniela Franco
Dr. Natarajan Ayithan



U01 HD092309

This publication was supported by funds through Maryland Department of Health's Cigarette Restitution Fund Program – CH-649-CRF and the National Cancer Institute – Cancer Center Support Grant (CCSG) – P30CA134274.d Service

Cytometry Discord

Bioconductor Zulip

https://github.com/DavidRach

@davidrach.bsky.social

www.linkedin.com/in/david-rach/

Want to collaborate? Reach out!

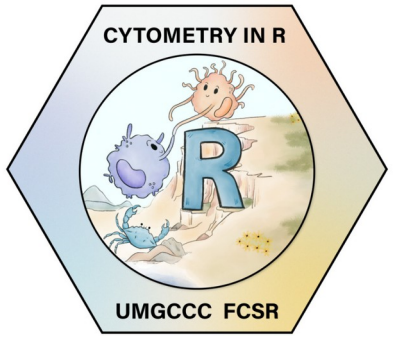
UNIVERSITY of MARYLAND
MARLENE AND STEWART GREENEBAUM
COMPREHENSIVE CANCER CENTER

ISAC RECOGNIZED
Shared Resource Laboratory

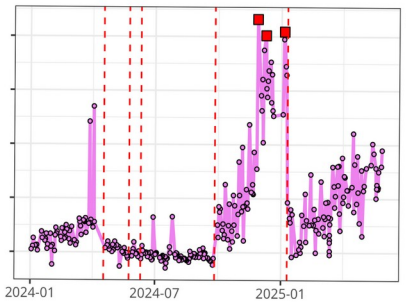
Questions?



Luciernaga



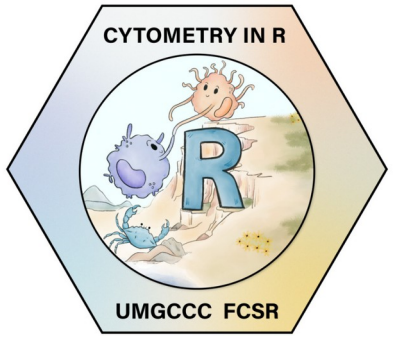
Website - CytometryInR



UMGCCC FCSR
InstrumentQC dashboard



Coereba



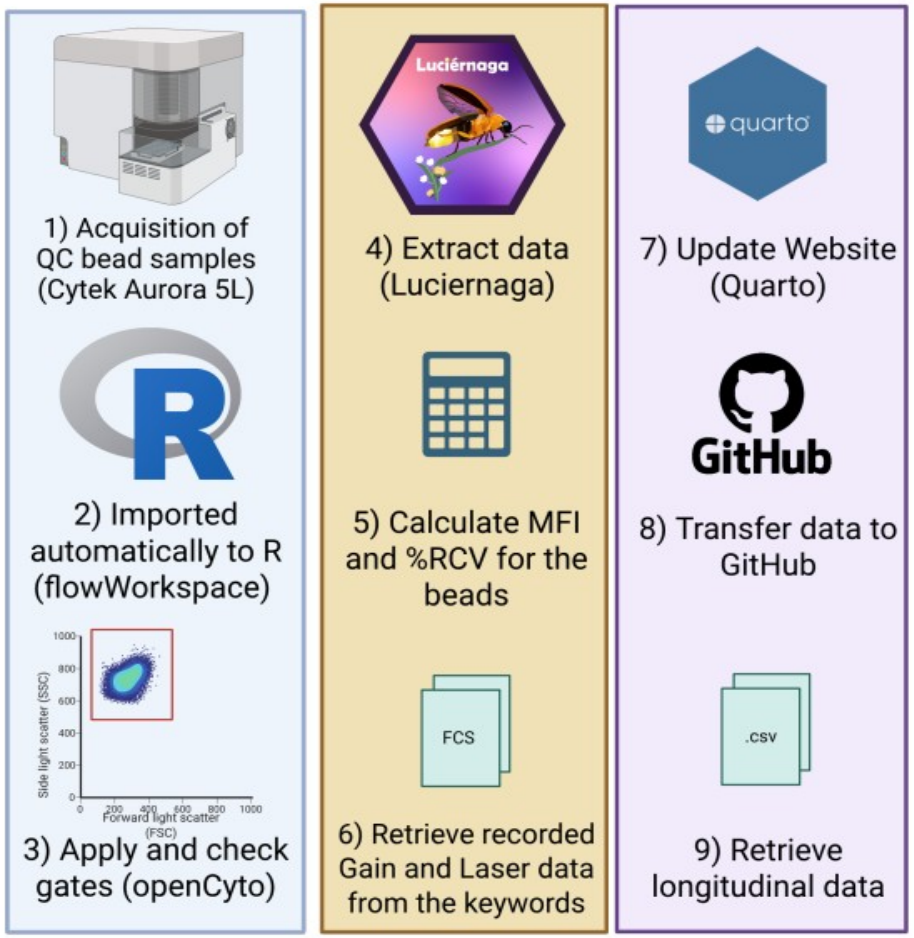
YouTube @**CytometryInR**



InstrumentQC How-To

CC BY SA

Automated Daily Processing





Behind-the-scenes:

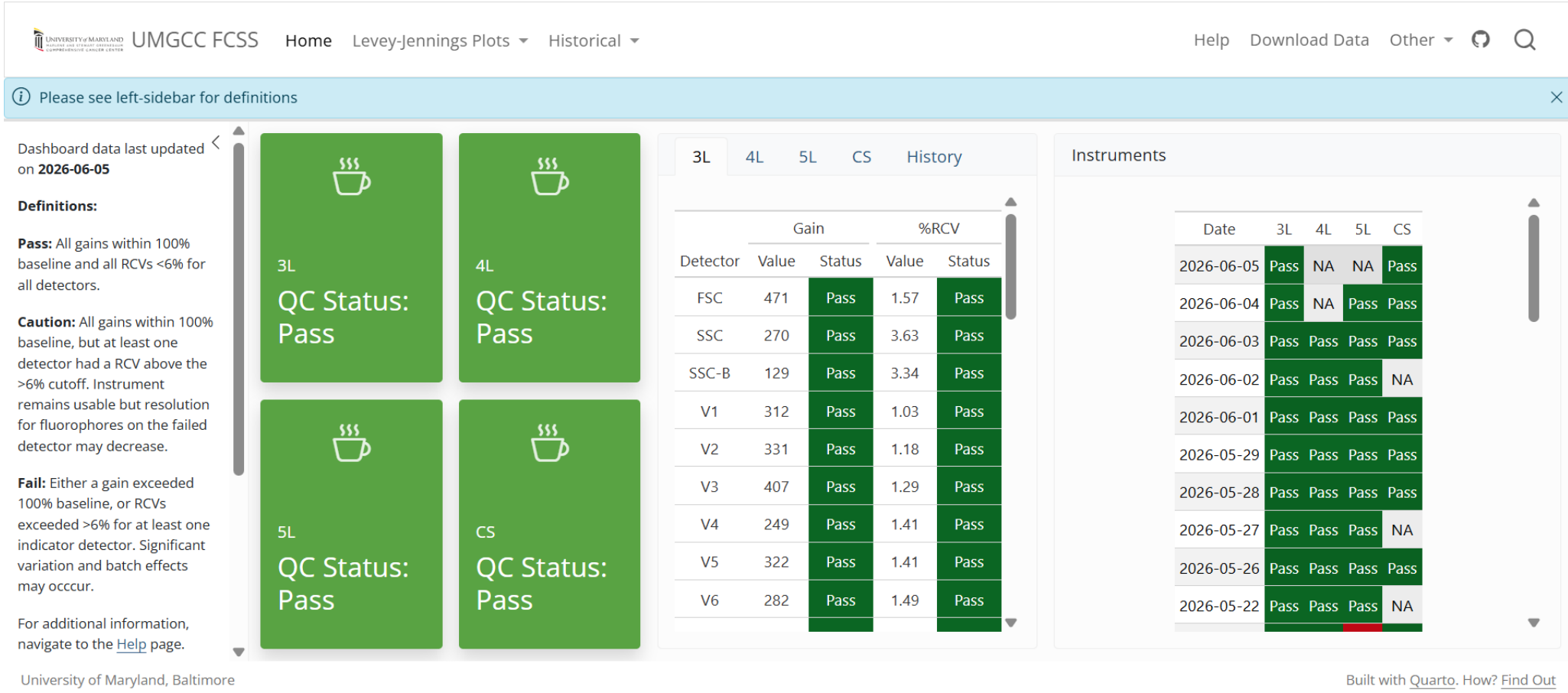
- Using R (and increasingly Rust) to process the .fcs files
- Automated transfer to GitHub via Windows Task Scheduler
- Our Quarto Website gets updated twice daily via a GitHub action

CC BY SA

Instrument Quality Control Dashboard

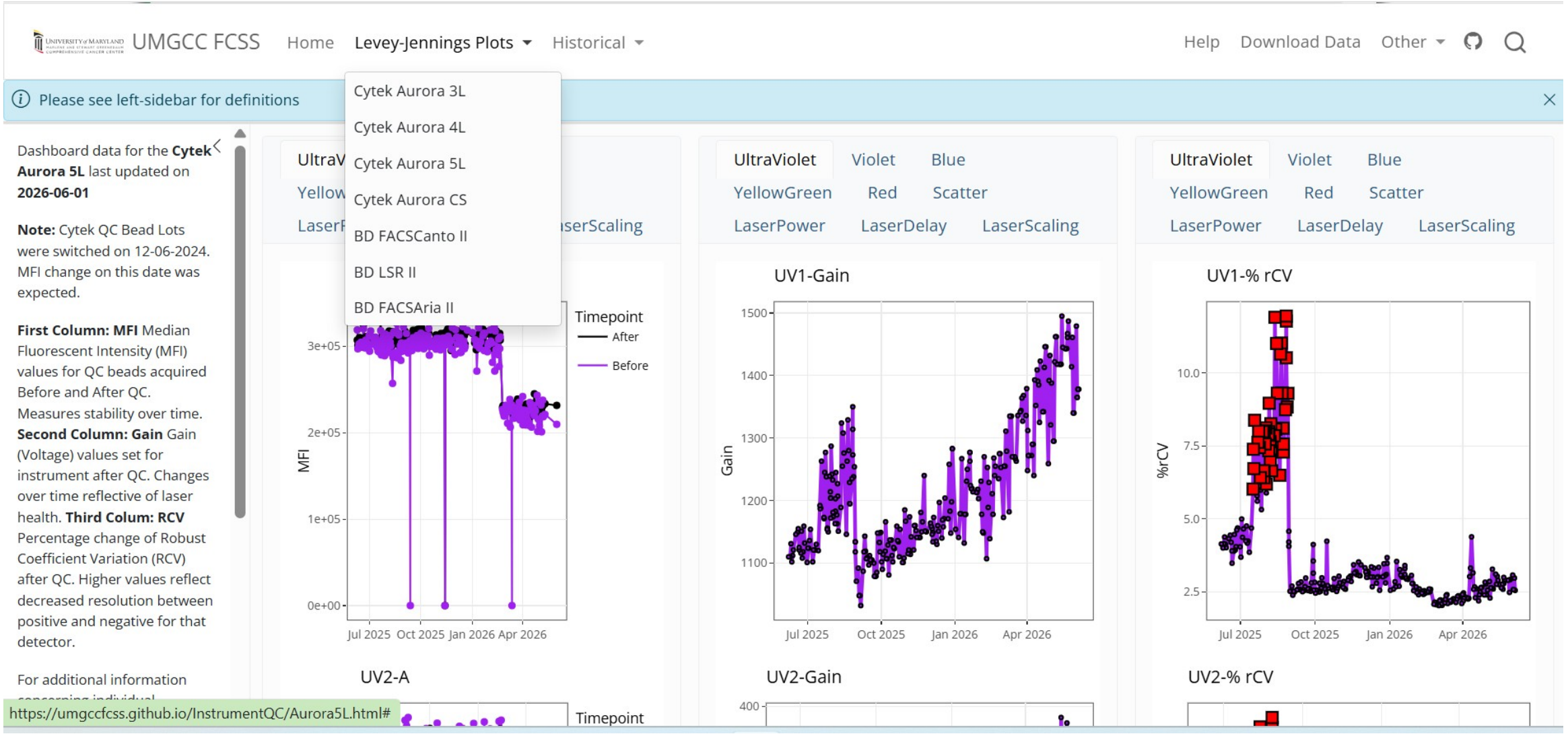
Dashboard

How-To



CC BY SA

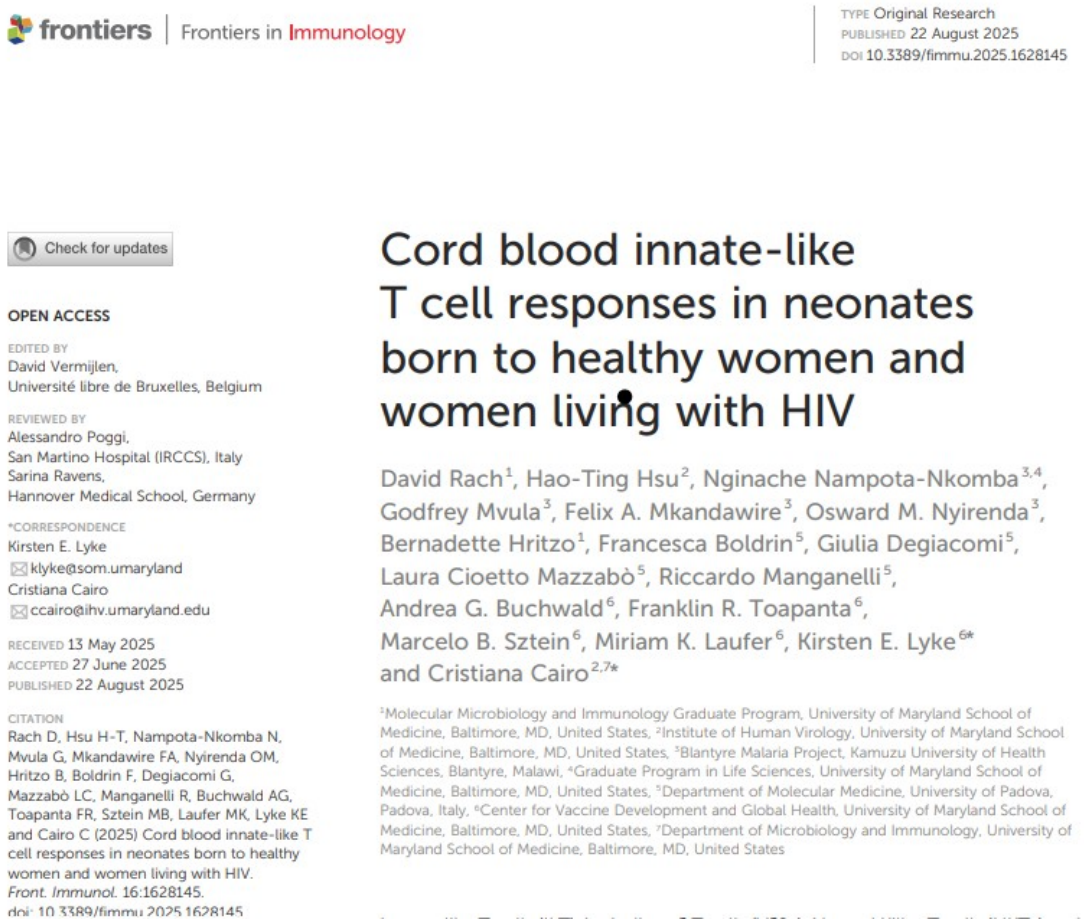
Instrument Quality Control Dashboard



CC BY SA

Published in Frontiers in Immunology

Paper



frontiers | Frontiers in Immunology

TYPE Original Research
PUBLISHED 22 August 2025
DOI 10.3389/fimmu.2025.1628145

Check for updates

OPEN ACCESS

EDITED BY
David Vermijlen,
Université libre de Bruxelles, Belgium

REVIEWED BY
Alessandro Poggi,
San Martino Hospital (IRCCS), Italy
Sarina Ravens,
Hannover Medical School, Germany

*CORRESPONDENCE
Kirsten E. Lyke
klyke@som.umaryland
Cristiana Cairo
ccairo@ihv.umaryland.edu

RECEIVED 13 May 2025
ACCEPTED 27 June 2025
PUBLISHED 22 August 2025

CITATION
Rach D, Hsu H-T, Nampota-Nkomba N, Mvula G, Mkandawire FA, Nyirenda OM, Hritzo B, Boldrin F, Degiacomi G, Mazzabò LC, Manganelli R, Buchwald AG, Toapanta FR, Sztein MB, Laufer MK, Lyke KE and Cairo C (2025) Cord blood innate-like T cell responses in neonates born to healthy women and women living with HIV. *Front. Immunol.* 16:1628145. doi: 10.3389/fimmu.2025.1628145

Cord blood innate-like T cell responses in neonates born to healthy women and women living with HIV

David Rach[1], Hao-Ting Hsu[2], Nginache Nampota-Nkomba[3,4], Godfrey Mvula[3], Felix A. Mkandawire[3], Osward M. Nyirenda[3], Bernadette Hritzo[1], Francesca Boldrin[5], Giulia Degiacomi[5], Laura Cioetto Mazzabò[5], Riccardo Manganelli[5], Andrea G. Buchwald[6], Franklin R. Toapanta[6], Marcelo B. Sztein[6], Miriam K. Laufer[6], Kirsten E. Lyke[6*] and Cristiana Cairo[2,7*]

[1]Molecular Microbiology and Immunology Graduate Program, University of Maryland School of Medicine, Baltimore, MD, United States, [2]Institute of Human Virology, University of Maryland School of Medicine, Baltimore, MD, United States, [3]Blantyre Malaria Project, Kamuzu University of Health Sciences, Blantyre, Malawi, [4]Graduate Program in Life Sciences, University of Maryland School of Medicine, Baltimore, MD, United States, [5]Department of Molecular Medicine, University of Padova, Padova, Italy, [6]Center for Vaccine Development and Global Health, University of Maryland School of Medicine, Baltimore, MD, United States, [7]Department of Microbiology and Immunology, University of Maryland School of Medicine, Baltimore, MD, United States

CC BY SA